

UPSC ISS Statistics Paper II

Statistical Inference Practice Bank (Q1–Q365)

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A sequential compilation of 355 multiple-choice questions (Q1–Q365) covering estimation theory, sufficiency, CRLB, MLE/UMVUE, confidence intervals, hypothesis testing (Neyman–Pearson, UMP), LRT/SPRT, and decision theory.

Note: question numbers [300, 301, 302, 303, 304, 305, 306, 307, 308, 309] were not present in the source material and are omitted.

Q1–Q40. Unbiasedness, Consistency, Efficiency, Sufficiency & Completeness — Foundations

Q1. Unbiasedness [MEDIUM]

An estimator $\hat{\theta}$ is unbiased if:

- **A)** $E[\hat{\theta}] = \theta$ for all θ
- **B)** $\text{Var}(\hat{\theta}) = 0$
- **C)** $\hat{\theta} \rightarrow \theta$ in probability
- **D)** $\text{MSE}(\hat{\theta}) = 0$

Answer: A

Trap/Note: C is consistency, not unbiasedness; D implies both bias=0 AND var=0 simultaneously

Q2. MSE Decomposition [MEDIUM]

$\text{MSE}(\hat{\theta})$ equals:

- **A)** $\text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$
- **B)** $\text{Var}(\hat{\theta}) - [\text{Bias}(\hat{\theta})]^2$
- **C)** $[\text{Bias}(\hat{\theta})]^2$ only
- **D)** $\text{Var}(\hat{\theta})$ only

Answer: A

Trap/Note: D holds only when estimator is unbiased; never subtract bias²

Q3. Consistency [MEDIUM]

$\hat{\theta}_n$ is consistent for θ if:

- A) $E[\hat{\theta}_n] \rightarrow \theta$ and $\text{Var}(\hat{\theta}_n) \rightarrow 0$
- B) $P(|\hat{\theta}_n - \theta| > \epsilon) \rightarrow 0$ for every $\epsilon > 0$
- C) $\hat{\theta}_n \rightarrow \theta$ almost surely
- D) Both A and B

Answer: B

Trap/Note: A is sufficient (by Chebyshev) but not necessary; C is strong consistency (stricter)

Q4. Efficiency [MEDIUM]

The relative efficiency of $\hat{\theta}_1$ w.r.t. $\hat{\theta}_2$ is:

- A) $\text{Var}(\hat{\theta}_1)/\text{Var}(\hat{\theta}_2)$
- B) $\text{Var}(\hat{\theta}_2)/\text{Var}(\hat{\theta}_1)$
- C) $\text{MSE}(\hat{\theta}_1)/\text{MSE}(\hat{\theta}_2)$
- D) $\text{Bias}(\hat{\theta}_1)/\text{Bias}(\hat{\theta}_2)$

Answer: B

Trap/Note: Efficiency of $\hat{\theta}_1 = \text{CRLB}/\text{Var}(\hat{\theta}_1)$; relative efficiency = Var of denominator over numerator

Q5. Sufficiency Definition [MEDIUM]

A statistic $T(X)$ is sufficient for θ if:

- A) T has minimum variance among all unbiased estimators
- B) The conditional distribution of X given T does not depend on θ
- C) T attains the Cramér-Rao lower bound
- D) T is a complete statistic

Answer: B

Trap/Note: Sufficiency is about conditional distribution, not variance; completeness is a separate concept

Q6. Admissibility [HIGH]

An estimator $\hat{\theta}$ is inadmissible if:

- A) It is biased
- B) There exists another estimator with smaller MSE for ALL θ
- C) It does not attain the CRLB
- D) It is not a function of sufficient statistic

Answer: B

Trap/Note: Biased estimators can be admissible (e.g., James-Stein); CRLB non-attainment $\neq$ inadmissibility

Q7. Ancillary Stat [HIGH]

A statistic $A(X)$ is ancillary if:

- **A)** Its distribution does not depend on θ
- **B)** It is independent of the sufficient statistic
- **C)** $E[A(X)] = 0$ for all θ
- **D)** It has minimum variance

Answer: A

Trap/Note: B is Basu's theorem consequence (ancillary $\perp$ complete sufficient), not the definition

Q8. Completeness Def [HIGH]

A statistic T is complete if:

- **A)** $E_{\theta}[g(T)] = 0$ for all θ implies $g(T) = 0$ a.s.
- **B)** T is sufficient and unbiased
- **C)** T achieves minimum variance
- **D)** The family of distributions of T is full-rank exponential

Answer: A

Trap/Note: D is sufficient for completeness (exponential family with full rank) but not the definition itself

Q9. Consistent Unbiased [MEDIUM]

Sample mean $\bar{X}$ from $N(\mu, \sigma^2)$ is:

- **A)** Unbiased and consistent for μ
- **B)** Consistent but biased for μ
- **C)** Unbiased but inconsistent for μ
- **D)** Neither unbiased nor consistent

Answer: A

Trap/Note: $E[\bar{X}] = \mu$ (unbiased); $\text{Var}(\bar{X}) = \sigma^2/n \rightarrow 0$ (consistent)

Q10. Best Linear Unbiased [MEDIUM]

By Gauss-Markov theorem, OLS estimator is BLUE meaning:

- **A)** Best (minimum variance) among all unbiased estimators
- **B)** Best (minimum variance) among all LINEAR unbiased estimators
- **C)** Best among all estimators including biased ones
- **D)** Minimum MSE among linear estimators

Answer: B

Trap/Note: “Linear” is crucial; MLE may be better overall; Gauss-Markov does NOT require normality

Q11. Asymptotic Efficiency [HIGH]

An estimator is asymptotically efficient if:

- **A)** $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1/I(\theta))$
- **B)** $\hat{\theta}$ is unbiased for all n
- **C)** $\text{Var}(\hat{\theta}) = 1/(nI(\theta))$ exactly
- **D)** $\hat{\theta}$ is consistent

Answer: A

Trap/Note: C is exact CRLB attainment (finite n); A is the asymptotic version; D alone is insufficient

Q12. Pitman Closeness [HIGH]

Pitman closeness criterion prefers $\hat{\theta}_1$ over $\hat{\theta}_2$ if:

- **A)** $\text{MSE}(\hat{\theta}_1) < \text{MSE}(\hat{\theta}_2)$
- **B)** $P(|\hat{\theta}_1 - \theta| < |\hat{\theta}_2 - \theta|) > 1/2$
- **C)** $\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2)$
- **D)** $\text{Bias}(\hat{\theta}_1) < \text{Bias}(\hat{\theta}_2)$

Answer: B

Trap/Note: Pitman closeness can contradict MSE ordering — classic ISS trap

Q13. Invariance MLE [MEDIUM]

If $\hat{\theta}$ is MLE of θ , then MLE of $g(\theta)$ is:

- **A)** $g(\hat{\theta})$ always
- **B)** $g(\hat{\theta})$ only if g is monotone
- **C)** $g(\hat{\theta})$ only if g is differentiable
- **D)** Obtained by separate maximization

Answer: A

Trap/Note: Invariance of MLE holds for ANY function g , not just monotone or differentiable ones

Q14. Unbiased Estimator Unique [HIGH]

If a complete sufficient statistic exists, then:

- **A)** There is at most one unbiased estimator based on it
- **B)** There is exactly one UMVUE
- **C)** All unbiased estimators are functions of it
- **D)** The MLE is always unbiased

Answer: B

Trap/Note: A is wrong (there can be many unbiased estimators); B follows from Lehmann-Scheffé theorem

Q15. Fisher Information Additive [MEDIUM]

For i.i.d. sample of size n , $I_n(\theta)$ equals:

- A) $I_1(\theta)/n$
- B) $n \cdot I_1(\theta)$
- C) $\sqrt{n} \cdot I_1(\theta)$
- D) $I_1(\theta)$

Answer: B

Trap/Note: Fisher information is ADDITIVE for independent observations; $\text{CRLB} = 1/(nI_1(\theta))$

Q16. Median Unbiased [HIGH]

An estimator $\hat{\theta}$ is median-unbiased if:

- A) $E[\hat{\theta}] = \theta$
- B) $P(\hat{\theta} \leq \theta) = P(\hat{\theta} \geq \theta) = 1/2$
- C) Median of $\hat{\theta}$ equals $E[\hat{\theta}]$
- D) $\hat{\theta}$ has symmetric distribution

Answer: B

Trap/Note: Mean-unbiasedness $\neq$ median-unbiasedness; for skewed distributions they differ

Q17. Cramer Rao Regularity [HIGH]

CRLB requires regularity conditions including:

- A) Support of $f(x; \theta)$ must NOT depend on θ
- B) θ must be a location parameter
- C) The distribution must be normal
- D) The sample size must be large

Answer: A

Trap/Note: For $U(0, \theta)$, $\text{Exp}(\theta)$ shifted — support depends on θ , so CRLB does NOT apply

Q18. Sufficiency Minimal [HIGH]

A minimal sufficient statistic is:

- A) A sufficient statistic of smallest possible dimension
- B) A function of every other sufficient statistic
- C) Both A and B
- D) The complete sufficient statistic

Answer: C

Trap/Note: Minimal sufficient = coarsest sufficient partition; complete sufficient $\subseteq$ minimal sufficient (when it exists)

Q19. Equivariant Estimator [HIGH]

A location equivariant estimator $\hat{\theta}$ satisfies:

- A) $\hat{\theta}(X + c) = \hat{\theta}(X) + c$ for all c
- B) $\hat{\theta}(cX) = c\hat{\theta}(X)$ for all $c > 0$
- C) $E[\hat{\theta}] = \theta$
- D) $\hat{\theta}$ is a function of order statistics

Answer: A

Trap/Note: B is scale equivariance; both together give affine equivariance

Q20. Consistent Estimator Function [MEDIUM]

If $\hat{\theta}_n \xrightarrow{p} \theta$ and g is continuous, then:

- A) $g(\hat{\theta}_n) \xrightarrow{p} g(\theta)$
- B) $g(\hat{\theta}_n) \xrightarrow{p} g(\theta)$ only if g is monotone
- C) $g(\hat{\theta}_n)$ may not be consistent
- D) $E[g(\hat{\theta}_n)] \rightarrow g(\theta)$

Answer: A

Trap/Note: Continuous mapping theorem guarantees A; D requires uniform integrability (stronger condition)

Q21. Bias Variance Tradeoff [MEDIUM]

Shrinkage estimators like James-Stein are preferred because:

- A) They are unbiased with smaller variance
- B) They introduce bias but reduce variance enough to lower total MSE
- C) They are consistent while OLS is not
- D) They satisfy CRLB

Answer: B

Trap/Note: James-Stein is BIASED; its advantage is total MSE reduction in $p \geq 3$ dimensions

Q22. Order Stat Distribution [MEDIUM]

PDF of k -th order statistic $X_{(k)}$ from $f(x)$ with CDF $F(x)$ is:

- A) $\frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x)$
- B) $\frac{n!}{k!(n-k)!} [F(x)]^k [1 - F(x)]^{n-k} f(x)$
- C) $n[F(x)]^{n-1} f(x)$
- D) $n[1 - F(x)]^{n-1} f(x)$

Answer: A

Trap/Note: C is PDF of maximum ($k = n$); D is PDF of minimum ($k = 1$); check factorial carefully

Q23. Exponential Family Complete [HIGH]

For a full-rank exponential family $f(x; \theta) = h(x) \exp[\eta(\theta)T(x) - B(\theta)]$:

- **A)** $T(x)$ is complete sufficient
- **B)** $T(x)$ is sufficient but not necessarily complete
- **C)** $T(x)$ is complete but not sufficient
- **D)** Completeness depends on sample size

Answer: A

Trap/Note: Full-rank (open set in natural parameter space) is the key condition; curved exponential families may lack completeness

Q24. Ancillary Basu [HIGH]

By Basu's theorem, if T is complete sufficient and A is ancillary, then:

- **A)** T and A are independent
- **B)** A is a function of T
- **C)** T and A have the same distribution
- **D)** A is also sufficient

Answer: A

Trap/Note: This is Basu's theorem — independence, not functional relationship; requires completeness

Q25. Unbiased Estimator Existence [HIGH]

For $U(0, \theta)$, an unbiased estimator of θ^{-1} based on $X_{(n)}$:

- **A)** Does not exist
- **B)** Is $\frac{n}{(n-1)X_{(n)}}$
- **C)** Is $\frac{1}{X_{(n)}}$
- **D)** Is $\frac{n+1}{nX_{(n)}}$

Answer: B

Trap/Note: $E[X_{(n)}] = \frac{n\theta}{n+1}$, so $E[1/X_{(n)}]$ requires careful computation; verify via $E[X_{(n)}^{-1}] = \frac{n}{(n-1)\theta}$

Q26. Lehmann Scheffe [HIGH]

Lehmann-Scheffé theorem states:

- **A)** Any unbiased estimator based on complete sufficient statistic is UMVUE
- **B)** UMVUE always attains CRLB
- **C)** Complete sufficient statistic is always MLE

- D) Rao-Blackwell improvement always yields UMVUE

Answer: A

Trap/Note: B is false (UMVUE may not attain CRLB); D requires completeness for uniqueness

Q27. Fisher Consistency [HIGH]

An estimator T_n is Fisher consistent if:

- A) $T_n \xrightarrow{p} \theta$ as $n \rightarrow \infty$
- B) The functional $T(F_\theta) = \theta$ where F_θ is the true CDF
- C) $E[T_n] = \theta$ for all n
- D) T_n is a function of sufficient statistic

Answer: B

Trap/Note: Fisher consistency is a property of the functional form, not a sequence property; different from ordinary consistency

Q28. Superefficiency [HIGH]

Hodges' superefficient estimator demonstrates:

- A) An estimator can have asymptotic variance below CRLB at isolated points
- B) MLE is never asymptotically efficient
- C) CRLB can be beaten uniformly
- D) Superefficiency implies admissibility

Answer: A

Trap/Note: Superefficiency occurs only on a set of measure zero; Le Cam showed it cannot hold on a dense set

Q29. Stein Paradox [HIGH]

James-Stein estimator dominates sample mean in terms of MSE when:

- A) $p \geq 1$ (any dimension)
- B) $p \geq 2$
- C) $p \geq 3$
- D) $p \geq 4$

Answer: C

Trap/Note: For $p = 1, 2$ sample mean is admissible; Stein paradox kicks in at $p \geq 3$

Q30. Moment Estimator Consistency [MEDIUM]

Method of moments estimators are generally:

- A) Consistent and efficient
- B) Consistent but not necessarily efficient
- C) Efficient but not consistent
- D) Neither consistent nor efficient

Answer: B

Trap/Note: MOM estimators are consistent (by LLN) but typically less efficient than MLE

Q31. CRLB Multiparameter [HIGH]

For multiparameter case, CRLB for $g(\theta)$ involves:

- A) Scalar Fisher information
- B) Fisher information matrix $I(\theta)$ and gradient ∇g
- C) Only diagonal elements of $I(\theta)$
- D) Determinant of $I(\theta)$

Answer: B

Trap/Note: $\text{Var}(\hat{g}) \geq [\nabla g]^T I(\theta)^{-1} [\nabla g]$; off-diagonal elements matter

Q32. Truncated Distribution MLE [HIGH]

For truncated normal $N(\mu, 1)$ truncated to $(0, \infty)$, MLE of μ :

- A) Is $\bar{X}$ (same as untruncated)
- B) Requires solving a nonlinear equation
- C) Does not exist
- D) Is the sample median

Answer: B

Trap/Note: Truncation changes the likelihood; MLE satisfies $\bar{X} = \mu + \phi(\mu)/\Phi(\mu)$ (Mills ratio equation)

Q33. Robust Estimator [MEDIUM]

The sample median is preferred over sample mean when:

- A) Distribution is symmetric
- B) Distribution has heavy tails or outliers
- C) Sample size is large
- D) Distribution is normal

Answer: B

Trap/Note: For normal distribution, mean is more efficient; median's advantage is robustness to outliers

Q34. Unbiased Variance Estimator [MEDIUM]

$S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ is unbiased for σ^2 because:

- A) $E[\sum (X_i - \bar{X})^2] = (n-1)\sigma^2$
- B) $E[\sum (X_i - \bar{X})^2] = n\sigma^2$
- C) $\bar{X}$ is independent of S^2 under normality
- D) Both A and C

Answer: A

Trap/Note: The $(n - 1)$ denominator corrects for one degree of freedom lost by estimating μ with $\bar{X}$

Q35. Jackknife Bias [MEDIUM]

The jackknife bias estimate for $\hat{\theta}$ is:

- A) $(n - 1)(\hat{\theta}_{(\cdot)} - \hat{\theta})$
- B) $n(\hat{\theta} - \hat{\theta}_{(\cdot)})$
- C) $\hat{\theta} - \hat{\theta}_{(\cdot)}$
- D) $(n - 1)(\hat{\theta} - \hat{\theta}_{(\cdot)})$

Answer: A

Trap/Note: $\hat{\theta}_{(\cdot)} = \frac{1}{n} \sum_i \hat{\theta}_{(i)}$ is mean of leave-one-out estimates; bias = $(n - 1)(\bar{\hat{\theta}}_{(\cdot)} - \hat{\theta})$

Q36. Bootstrap Principle [MEDIUM]

Bootstrap estimates sampling distribution of $\hat{\theta}$ by:

- A) Analytical approximation using CLT
- B) Resampling WITH replacement from observed data
- C) Resampling WITHOUT replacement from observed data
- D) Generating data from fitted parametric model only

Answer: B

Trap/Note: With replacement is key; without replacement is subsampling (different method); parametric bootstrap uses fitted model

Q37. Delta Method [MEDIUM]

By delta method, asymptotic variance of $g(\bar{X})$ is:

- A) $[g'(\mu)]^2 \cdot \sigma^2/n$
- B) $g'(\mu) \cdot \sigma^2/n$
- C) $[g(\mu)]^2 \cdot \sigma^2/n$
- D) $[g''(\mu)]^2 \cdot \sigma^2/n$

Answer: A

Trap/Note: First derivative squared, not second; applies when $g'(\mu) \neq 0$

Q38. Variance Stabilizing [HIGH]

Variance stabilizing transformation for $\text{Poisson}(\lambda)$ is:

- A) $\sqrt{X}$
- B) $\log X$
- C) X^2
- D) $1/X$

Answer: A

Trap/Note: For Poisson, $\text{Var}(X) = \lambda = E[X]$; VST satisfies $g'(\mu) \propto 1/\sqrt{\mu}$, giving $g(\mu) = 2\sqrt{\mu}$; $\sqrt{X}$ is the answer

Q39. Ancillary Example [HIGH]

For location family $f(x - \theta)$, which is ancillary?

- A) Sample mean $\bar{X}$
- B) Sample range $X_{(n)} - X_{(1)}$
- C) Sample maximum $X_{(n)}$
- D) Sample variance S^2 (when θ is location only)

Answer: B

Trap/Note: Range and all spacings are ancillary for location families; S^2 is ancillary only if scale is known

Q40. Completeness Bernoulli [MEDIUM]

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, $T = \sum X_i$ is complete because:

- A) It is sufficient
- B) $E_p[g(T)] = 0$ for all $p \in (0, 1)$ implies $g \equiv 0$ (polynomial argument)
- C) It is the MLE
- D) It has binomial distribution

Answer: B

Trap/Note: Completeness proof: $E[g(T)] = \sum_{t=0}^n g(t) \binom{n}{t} p^t (1-p)^{n-t} = 0$ for all p implies all $g(t) = 0$

Q41–Q80. Cramér–Rao Lower Bound & Fisher Information

Q41. CRLB Formula [MEDIUM]

For unbiased estimator $\hat{\theta}$, CRLB states:

- A) $\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$
- B) $\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$
- C) $\text{Var}(\hat{\theta}) \geq \frac{n}{I(\theta)}$
- D) $\text{Var}(\hat{\theta}) \geq I(\theta)$

Answer: A

Trap/Note: For i.i.d. sample of size n : $\text{CRLB} = 1/(nI_1(\theta))$ where I_1 is single-observation Fisher info

Q42. Fisher Info Normal [MEDIUM]

For $X \sim N(\mu, \sigma^2)$ with σ^2 known, $I(\mu)$ equals:

- A) $1/\sigma^2$
- B) σ^2
- C) $2/\sigma^4$
- D) $1/(2\sigma^2)$

Answer: A

Trap/Note: $\log f = -\frac{(x-\mu)^2}{2\sigma^2} + C$; $\partial^2 \log f / \partial \mu^2 = -1/\sigma^2$; $I(\mu) = 1/\sigma^2$

Q43. Fisher Info Bernoulli [MEDIUM]

For $X \sim \text{Bernoulli}(p)$, $I(p)$ equals:

- A) $p(1-p)$
- B) $1/[p(1-p)]$
- C) $p/(1-p)$
- D) $1/p + 1/(1-p)$

Answer: B

Trap/Note: $I(p) = E[(\partial \log f / \partial p)^2] = 1/[p(1-p)]$; CRLB for $\bar{X}$ is $p(1-p)/n$ — verify: $\text{Var}(\bar{X}) = p(1-p)/n$ ✓ attains CRLB

Q44. Fisher Info Poisson [MEDIUM]

For $X \sim \text{Poisson}(\lambda)$, $I(\lambda)$ equals:

- A) λ
- B) $1/\lambda$
- C) λ^2
- D) $1/\lambda^2$

Answer: B

Trap/Note: $\log f = x \log \lambda - \lambda - \log(x!)$; $\partial^2 \log f / \partial \lambda^2 = -x/\lambda^2$; $I(\lambda) = E[X]/\lambda^2 = 1/\lambda$

Q45. CRLB Function Parameter [MEDIUM]

CRLB for estimating $g(\theta)$ (unbiased) is:

- A) $\frac{[g'(\theta)]^2}{nI(\theta)}$
- B) $\frac{g'(\theta)}{nI(\theta)}$
- C) $\frac{[g(\theta)]^2}{nI(\theta)}$
- D) $\frac{[g'(\theta)]^2}{I(\theta)}$

Answer: A

Trap/Note: Chain rule applies: $[g'(\theta)]^2$ in numerator; D forgets the n factor

Q46. CRLB Attainment Condition [HIGH]

CRLB is attained if and only if:

- A) $\frac{\partial \log f}{\partial \theta} = A(\theta)[T(x) - g(\theta)]$ for some functions A, g
- B) The distribution is normal
- C) The estimator is MLE
- D) The sample size is large

Answer: A

Trap/Note: This is the exact attainment condition; it implies $T(x)$ is UMVUE for $g(\theta)$; only exponential family members typically satisfy this

Q47. CRLB Exponential [MEDIUM]

For $X \sim \text{Exp}(\theta)$ (rate θ), CRLB for $\hat{\theta}$ (unbiased) is:

- A) θ^2/n
- B) θ/n
- C) $1/(n\theta^2)$
- D) n/θ^2

Answer: A

Trap/Note: $I(\theta) = 1/\theta^2$; CRLB = θ^2/n ; MLE $1/\bar{X}$ attains this bound

Q48. CRLB Uniform DNE [HIGH]

For $X \sim U(0, \theta)$, CRLB for θ :

- A) Equals $\theta^2/(3n)$
- B) Equals θ^2/n
- C) Does not exist
- D) Equals $4\theta^2/n$

Answer: C

Trap/Note: Support $(0, \theta)$ depends on θ ; regularity condition violated; CRLB framework inapplicable; UMVUE variance is $\theta^2/[n(n+2)]$

Q49. Information Inequality Biased [HIGH]

For biased estimator with bias $b(\theta)$, the generalized CRLB is:

- A) $\text{Var}(\hat{\theta}) \geq \frac{[1+b'(\theta)]^2}{nI(\theta)}$
- B) $\text{Var}(\hat{\theta}) \geq \frac{[b'(\theta)]^2}{nI(\theta)}$
- C) $\text{MSE}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$
- D) $\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$

Answer: A

Trap/Note: $[1 + b'(\theta)]^2$ not $[b'(\theta)]^2$; when $b(\theta) = 0$ (unbiased), reduces to standard CRLB

Q50. Bhattacharya Bound [HIGH]

Bhattacharya bounds improve upon CRLB by:

- A) Using higher-order score functions (derivatives of log-likelihood)
- B) Using larger sample sizes
- C) Restricting to exponential families
- D) Using sufficient statistics

Answer: A

Trap/Note: Bhattacharya uses $\partial^k \log f / \partial \theta^k$ for $k = 1, 2, \dots$; first-order Bhattacharya = CRLB; higher orders give tighter bounds

Q51. Bhattacharya Normal Theta Sq [HIGH]

For $X \sim N(\theta, 1)$, estimating θ^2 , Bhattacharya bound (2nd order) is:

- A) $4\theta^2/n$
- B) $4\theta^2/n + 2/n^2$
- C) $2/n$
- D) $4\theta^2/n - 2/n^2$

Answer: B

Trap/Note: CRLB gives $4\theta^2/n$; 2nd-order Bhattacharya adds $2/n^2$ term; UMVUE $\bar{X}^2 - 1/n$ has variance $4\theta^2/n + 2/n^2$ — attains 2nd-order bound

Q52. Fisher Info Gamma [HIGH]

For $X \sim \Gamma(\alpha, \beta)$ with α known, $I(\beta)$ equals:

- A) α/β^2
- B) β^2/α
- C) $1/(\alpha\beta^2)$
- D) α^2/β^2

Answer: A

Trap/Note: $E[X] = \alpha\beta$, $\text{Var}(X) = \alpha\beta^2$; $I(\beta) = \alpha/\beta^2$; CRLB for $\hat{\beta} = \bar{X}/\alpha$ is $\beta^2/(n\alpha)$ ✓

Q53. CRLB Location Family [HIGH]

For location family $f(x - \theta)$, Fisher information is:

- A) $I(\theta) = \int \left[\frac{f'(x)}{f(x)} \right]^2 f(x) dx$ (independent of θ)
- B) $I(\theta) = \theta^2 \int \left[\frac{f'(x)}{f(x)} \right]^2 f(x) dx$
- C) $I(\theta)$ depends on θ
- D) $I(\theta) = 1/\sigma^2$ always

Answer: A

Trap/Note: For location families, Fisher information is constant (does not depend on θ); this is a key property

Q54. CRLB Scale Family [HIGH]

For scale family $\frac{1}{\theta} f(x/\theta)$, $I(\theta)$ equals:

- A) $\frac{1}{\theta^2} \int \left[1 + \frac{x f'(x)}{f(x)} \right]^2 f(x) dx$
- B) $\frac{1}{\theta^2}$ always
- C) $\theta^2 \int \left[\frac{f'(x)}{f(x)} \right]^2 f(x) dx$
- D) $1/\theta^2$ times location Fisher information

Answer: A

Trap/Note: Score for scale family: $\partial \log f / \partial \theta = -1/\theta - (x/\theta^2) f'(x/\theta)/f(x/\theta)$; careful substitution needed

Q55. Efficient Estimator Definition [MEDIUM]

An unbiased estimator is efficient if:

- A) Its variance equals CRLB
- B) Its variance is less than all other unbiased estimators
- C) Both A and B (they are equivalent when CRLB is attained)
- D) It is consistent

Answer: C

Trap/Note: When CRLB is attained, the estimator is UMVUE; A and B are equivalent in this context

Q56. CRLB Cauchy [HIGH]

For $X \sim \text{Cauchy}(\theta, 1)$, $I(\theta)$ equals:

- A) $1/2$
- B) 2
- C) 1
- D) Does not exist

Answer: A

Trap/Note: $f(x; \theta) = \frac{1}{\pi[1+(x-\theta)^2]}$; $I(\theta) = \int \frac{4(x-\theta)^2}{\pi^2[1+(x-\theta)^2]^4} \cdot \pi[1+(x-\theta)^2]dx = 1/2$; sample mean is inconsistent for Cauchy

Q57. Information Matrix Bivariate [HIGH]

For $N(\mu, \sigma^2)$ with both parameters unknown, Fisher information matrix is:

- A) $\text{diag}(1/\sigma^2, 1/(2\sigma^4))$
- B) $\text{diag}(n/\sigma^2, n/(2\sigma^4))$
- C) $\text{diag}(1/\sigma^2, 2\sigma^4)$
- D) Non-diagonal matrix

Answer: A

Trap/Note: Off-diagonal is zero (score functions are uncorrelated); single-observation matrix; multiply by n for sample

Q58. CRLB Beta Distribution [HIGH]

For $X \sim \text{Beta}(\theta, 1)$, $f(x; \theta) = \theta x^{\theta-1}$, $I(\theta)$ equals:

- A) $1/\theta^2$
- B) θ^2
- C) $1/\theta$
- D) θ

Answer: A

Trap/Note: $\log f = \log \theta + (\theta - 1) \log x$; $\partial \log f / \partial \theta = 1/\theta + \log x$; $I(\theta) = \text{Var}(1/\theta + \log X) = \text{Var}(\log X) = 1/\theta^2$ (since $-\log X \sim \text{Exp}(\theta)$)

Q59. CRLB Truncated [HIGH]

For $X \sim \text{Exp}(\theta)$ truncated to $(0, c)$, CRLB:

- A) Is same as untruncated: θ^2/n
- B) Is smaller than θ^2/n (more information)
- C) Is larger than θ^2/n (less information)
- D) Does not exist

Answer: B

Trap/Note: Truncation adds information (we know $X < c$); Fisher information increases; CRLB decreases

Q60. Bhattacharya First Order [MEDIUM]

The first-order Bhattacharya bound coincides with:

- A) CRLB
- B) Rao-Blackwell bound
- C) Chapman-Robbins bound
- D) Hammersley-Chapman-Robbins bound

Answer: A

Trap/Note: Bhattacharya hierarchy: 1st order = CRLB; higher orders give tighter bounds; Chapman-Robbins is a different (non-differentiability-requiring) bound

Q61. Fisher Info Geometric [MEDIUM]

For $X \sim \text{Geometric}(p)$, $P(X = k) = (1 - p)^{k-1}p$, $I(p)$ equals:

- A) $\frac{1}{p^2(1-p)}$
- B) $\frac{1}{p(1-p)}$
- C) $\frac{1}{p^2}$
- D) $\frac{1}{(1-p)^2}$

Answer: A

Trap/Note: $E[X] = 1/p$; $\text{Var}(X) = (1 - p)/p^2$; $I(p) = 1/[p^2(1 - p)]$; CRLB for $\hat{p} = 1/\bar{X}$ is $p^2(1 - p)/n$

Q62. CRLB Negative Binomial [HIGH]

For $X \sim \text{NegBin}(r, p)$ with r known, $I(p)$ equals:

- A) $\frac{r}{p^2(1-p)}$
- B) $\frac{r}{p(1-p)}$
- C) $\frac{1}{p^2(1-p)}$
- D) $\frac{r^2}{p^2(1-p)}$

Answer: A

Trap/Note: NegBin is exponential family; $I(p) = r/[p^2(1 - p)]$; for $r = 1$ reduces to geometric case

Q63. Attainment Exponential Family [HIGH]

In exponential family $f(x; \theta) = h(x) \exp[\eta(\theta)T(x) - B(\theta)]$, CRLB for $E_\theta[T(X)]$ is attained by:

- A) $T(X)$ itself
- B) $\bar{T} = \frac{1}{n} \sum T(X_i)$
- C) MLE of $E_\theta[T(X)]$
- D) All of the above

Answer: D

Trap/Note: In exponential family, $\bar{T}$ is UMVUE for $E[T]$, equals MLE (by invariance), and attains CRLB — all three coincide

Q64. CRLB Double Exponential [HIGH]

For $X \sim \text{Laplace}(\theta, 1)$, $f(x; \theta) = \frac{1}{2}e^{-|x-\theta|}$, $I(\theta)$ equals:

- A) 1
- B) $1/2$

- C) 2
- D) Does not exist

Answer: A

Trap/Note: $\partial \log f / \partial \theta = \text{sgn}(x - \theta)$; $I(\theta) = E[\text{sgn}(X - \theta)^2] = 1$; sample median (not mean) is MLE and UMVUE

Q65. CRLB Weibull [HIGH]

For Weibull with shape k known and scale λ : $f(x; \lambda) = \frac{k}{\lambda} (x/\lambda)^{k-1} e^{-(x/\lambda)^k}$, $I(\lambda)$ equals:

- A) k^2/λ^2
- B) k/λ^2
- C) $1/\lambda^2$
- D) k^2/λ

Answer: A

Trap/Note: Score: $\partial \log f / \partial \lambda = -k/\lambda + k(x/\lambda)^k/\lambda$; $I(\lambda) = k^2/\lambda^2$; CRLB = $\lambda^2/(nk^2)$

Q66. Information Reparametrization [HIGH]

If $\phi = g(\theta)$ and g is differentiable, then $I_\phi(\phi)$ equals:

- A) $I_\theta(\theta)/[g'(\theta)]^2$
- B) $I_\theta(\theta) \cdot [g'(\theta)]^2$
- C) $I_\theta(\theta) \cdot g'(\theta)$
- D) $I_\theta(\theta)/g'(\theta)$

Answer: A

Trap/Note: Fisher information transforms inversely to variance; CRLB for $g(\theta)$ is $[g'(\theta)]^2/(nI_\theta)$ which equals $1/(nI_\phi)$ ✓

Q67. CRLB Mixture [HIGH]

For mixture $f(x; \theta) = \theta f_1(x) + (1 - \theta) f_2(x)$, $I(\theta)$ equals:

- A) $\int \frac{[f_1(x) - f_2(x)]^2}{\theta f_1(x) + (1 - \theta) f_2(x)} dx$
- B) $\theta I_1(\theta) + (1 - \theta) I_2(\theta)$
- C) $I_1(\theta) + I_2(\theta)$
- D) $\theta^2 I_1(\theta) + (1 - \theta)^2 I_2(\theta)$

Answer: A

Trap/Note: Score = $[f_1(x) - f_2(x)]/f(x; \theta)$; $I(\theta) = E[\text{score}^2]$ gives A; B is wrong — mixture Fisher info is NOT weighted average

Q68. CRLB Censored [HIGH]

For Type-I right-censored exponential data, Fisher information compared to complete data:

- A) Is larger (censoring adds information about censoring mechanism)
- B) Is smaller (censoring loses information)

- C) Is the same
- D) Depends on censoring time

Answer: B

Trap/Note: Censoring always loses information; $I_{\text{censored}} < I_{\text{complete}}$; CRLB is larger for censored data

Q69. Bhattacharya Poisson [HIGH]

For $X \sim \text{Poisson}(\lambda)$, estimating $e^{-2\lambda}$, CRLB equals:

- A) $4\lambda e^{-4\lambda}/n$
- B) $4e^{-4\lambda}/n$
- C) $4\lambda^2 e^{-4\lambda}/n$
- D) $e^{-4\lambda}/n$

Answer: A

Trap/Note: $g(\lambda) = e^{-2\lambda}$; $g'(\lambda) = -2e^{-2\lambda}$; $I(\lambda) = 1/\lambda$; $\text{CRLB} = [g'(\lambda)]^2 / (nI(\lambda)) = 4e^{-4\lambda} \cdot \lambda/n$

Q70. CRLB Pareto [HIGH]

For Pareto $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, $I(\theta)$ equals:

- A) $1/\theta^2$
- B) θ^2
- C) $1/\theta$
- D) θ

Answer: A

Trap/Note: $\log f = \log \theta - (\theta + 1) \log x$; $\partial \log f / \partial \theta = 1/\theta - \log x$; $I(\theta) = \text{Var}(\log X) = 1/\theta^2$ (since $\log X \sim \text{Exp}(\theta)$)

Q71. CRLB Inverse Gaussian [HIGH]

For Inverse Gaussian $IG(\mu, \lambda)$ with λ known, $I(\mu)$ equals:

- A) λ/μ^3
- B) μ^3/λ
- C) $1/\mu^2$
- D) λ^2/μ^3

Answer: A

Trap/Note: $I(\mu) = \lambda/\mu^3$; CRLB for $\hat{\mu} = \bar{X}$ is $\mu^3/(n\lambda)$; verify: $\text{Var}(\bar{X}) = \mu^3/(n\lambda)$ ✓ attains CRLB

Q72. CRLB Logistic [HIGH]

For $X \sim \text{Logistic}(\theta, 1)$, $I(\theta)$ equals:

- A) $1/3$
- B) 3

- C) $\pi^2/3$
- D) $3/\pi^2$

Answer: A

Trap/Note: $f(x; \theta) = e^{-(x-\theta)}/[1+e^{-(x-\theta)}]^2$; $I(\theta) = 1/3$; CRLB = $3/n$; sample mean has variance $\pi^2/(3n) > 3/n$ — not efficient

Q73. CRLB Binomial P [MEDIUM]

For $X \sim \text{Bin}(n, p)$ (single observation), $I(p)$ equals:

- A) $n/[p(1-p)]$
- B) $1/[p(1-p)]$
- C) $np(1-p)$
- D) $p(1-p)/n$

Answer: A

Trap/Note: $\text{Binomial}(n, p) = \text{sum of } n \text{ Bernoulli}(p)$; $I_{\text{Bin}(n,p)} = n \cdot I_{\text{Bern}(p)} = n/[p(1-p)]$

Q74. CRLB Multivariate Normal [HIGH]

For $X \sim N_p(\mu, \Sigma)$ with Σ known, $I(\mu)$ (as matrix) equals:

- A) Σ^{-1}
- B) Σ
- C) $|\Sigma|^{-1}I_p$
- D) $\text{tr}(\Sigma^{-1})I_p$

Answer: A

Trap/Note: Fisher information matrix for location in multivariate normal is Σ^{-1} ; CRLB matrix is Σ/n ; $\bar{X}$ attains it

Q75. CRLB Curved Exponential [HIGH]

For curved exponential family (e.g., $N(\theta, \theta^2)$), CRLB:

- A) Is attained by MLE asymptotically
- B) Is never attained exactly for finite n
- C) Does not exist
- D) Is attained by method of moments

Answer: B

Trap/Note: Curved exponential families generally cannot attain CRLB for finite n ; MLE is asymptotically efficient

Q76. Information Processing Inequality [HIGH]

Data processing inequality for Fisher information states:

- A) $I_T(\theta) \leq I_X(\theta)$ for any statistic $T = T(X)$
- B) $I_T(\theta) \geq I_X(\theta)$ for sufficient T

- C) $I_T(\theta) = I_X(\theta)$ always
- D) $I_T(\theta) = I_X(\theta)$ if and only if T is sufficient

Answer: D

Trap/Note: A is true in general; D is the equality condition (sufficiency); B is wrong — sufficient statistic preserves, not increases, information

Q77. CRLB Sequential [HIGH]

In sequential sampling with random sample size N , CRLB:

- A) Is $1/(E[N] \cdot I(\theta))$
- B) Is $1/(N \cdot I(\theta))$ with N fixed
- C) Does not apply
- D) Requires Wald's identity

Answer: A

Trap/Note: For sequential procedures, expected Fisher information is $E[N] \cdot I_1(\theta)$; CRLB uses expected sample size

Q78. Bhattacharya Normal Variance [HIGH]

For $X \sim N(0, \sigma^2)$, estimating σ^2 , CRLB equals:

- A) $2\sigma^4/n$
- B) σ^4/n
- C) $4\sigma^4/n$
- D) σ^2/n

Answer: A

Trap/Note: $I(\sigma^2) = 1/(2\sigma^4)$; CRLB = $2\sigma^4/n$; $\frac{1}{n} \sum X_i^2$ is UMVUE and attains CRLB (since $\mu = 0$ known)

Q79. CRLB Attained Examples [HIGH]

Which distribution's MLE does NOT attain CRLB for finite n ?

- A) $N(\mu, \sigma^2)$ with σ^2 known, estimating μ
- B) $\text{Poisson}(\lambda)$, estimating λ
- C) $N(\mu, \sigma^2)$ with both unknown, estimating σ^2
- D) $\text{Bernoulli}(p)$, estimating p

Answer: C

Trap/Note: S^2 (with $n - 1$ denominator) is UMVUE but does NOT attain CRLB; MLE $\hat{\sigma}^2$ (with n denominator) is biased; neither attains CRLB

Q80. CRLB Regularity Summary [HIGH]

Which condition is NOT required for CRLB?

- A) Support of $f(x; \theta)$ independent of θ

- **B)** $f(x; \theta)$ twice differentiable in θ
- **C)** Interchange of differentiation and integration
- **D)** Distribution must be from exponential family

Answer: D

Trap/Note: CRLB applies to any regular family; exponential family is sufficient for attainment but NOT required for the bound itself

Q81–Q110. Factorization Theorem, Sufficiency & Completeness — Worked Families

Q81. Factorization Theorem [MEDIUM]

By Fisher-Neyman factorization theorem, $T(X)$ is sufficient for θ iff:

- A) $f(x; \theta) = g(T(x); \theta) \cdot h(x)$
- B) $f(x; \theta) = g(T(x)) \cdot h(x; \theta)$
- C) $f(x; \theta) = g(x) \cdot h(T(x); \theta)$
- D) $f(x; \theta) = g(T(x); \theta) + h(x)$

Answer: A

Trap/Note: g depends on x ONLY through $T(x)$ and on θ ; h depends on x but NOT on θ ; product not sum

Q82. Sufficient Stat Normal [MEDIUM]

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with both unknown, minimal sufficient statistic is:

- A) $(\bar{X}, S^2)$
- B) $(\bar{X}, \sum X_i^2)$
- C) $(\sum X_i, \sum X_i^2)$
- D) All of the above are equivalent

Answer: D

Trap/Note: All three are equivalent (one-to-one functions of each other); minimal sufficient statistic is not unique in representation

Q83. Sufficient Stat Uniform [MEDIUM]

For $X_1, \dots, X_n \sim U(0, \theta)$, sufficient statistic is:

- A) $\bar{X}$
- B) $X_{(n)}$ (maximum)
- C) $X_{(1)}$ (minimum)
- D) $\sum X_i$

Answer: B

Trap/Note: $f(x; \theta) = \theta^{-n} I(X_{(n)} \leq \theta)$; factorization gives $T = X_{(n)}$; $\bar{X}$ is NOT sufficient here

Q84. Sufficient Stat Exponential [MEDIUM]

For $X_1, \dots, X_n \sim \text{Exp}(\theta)$, sufficient statistic is:

- A) $X_{(1)}$
- B) $\bar{X}$
- C) $\sum X_i$
- D) Both B and C

Answer: D

Trap/Note: $\bar{X}$ and $\sum X_i$ are one-to-one functions of each other; both are sufficient; $X_{(1)}$ is NOT sufficient for rate parameter

Q85. Sufficient Stat Cauchy [HIGH]

For $X_1, \dots, X_n \sim \text{Cauchy}(\theta, 1)$, sufficient statistic is:

- A) $\bar{X}$
- B) Sample median
- C) $(X_{(1)}, X_{(2)}, \dots, X_{(n)})$ — all order statistics
- D) $X_{(1)}$ and $X_{(n)}$

Answer: C

Trap/Note: Cauchy is NOT exponential family; likelihood does not factor to reduce dimension; full order statistics are minimal sufficient

Q86. Complete Sufficient Uniform [HIGH]

For $U(0, \theta)$, $X_{(n)}$ is complete sufficient. UMVUE of θ is:

- A) $X_{(n)}$
- B) $\frac{n+1}{n}X_{(n)}$
- C) $2\bar{X}$
- D) $\frac{n}{n+1}X_{(n)}$

Answer: B

Trap/Note: $E[X_{(n)}] = \frac{n\theta}{n+1}$; unbiased correction: $\frac{n+1}{n}X_{(n)}$; $2\bar{X}$ is unbiased but NOT UMVUE (not function of complete sufficient stat)

Q87. Factorization Beta [MEDIUM]

For $X_1, \dots, X_n \sim \text{Beta}(\theta, 1)$, $f(x; \theta) = \theta x^{\theta-1}$, sufficient statistic is:

- A) $\sum X_i$
- B) $\prod X_i$
- C) $\sum \log X_i$
- D) Both B and C

Answer: D

Trap/Note: $\log f = n \log \theta + (\theta - 1) \sum \log x_i$; $T = \sum \log X_i$ is sufficient; $\prod X_i = \exp(\sum \log X_i)$ is equivalent

Q88. Complete Stat Uniform Both [HIGH]

For $X_1, \dots, X_n \sim U(\alpha, \beta)$ with both unknown, complete sufficient statistic is:

- A) $(X_{(1)}, X_{(n)})$
- B) $(\bar{X}, S^2)$
- C) $X_{(1)}$ alone
- D) $X_{(n)}$ alone

Answer: A

Trap/Note: Both endpoints needed; $(X_{(1)}, X_{(n)})$ is complete sufficient for (α, β) ; neither alone is sufficient

Q89. Factorization Shifted Exp [HIGH]

For $X_1, \dots, X_n \sim f(x; \theta) = e^{-(x-\theta)}$, $x > \theta$, sufficient statistic is:

- A) $\sum X_i$
- B) $\bar{X}$
- C) $X_{(1)}$ (minimum)
- D) $X_{(n)}$ (maximum)

Answer: C

Trap/Note: $f(\mathbf{x}; \theta) = e^{-\sum(x_i - \theta)} I(X_{(1)} > \theta) = e^{n\theta} e^{-\sum x_i} I(X_{(1)} > \theta)$; $T = X_{(1)}$ is sufficient

Q90. Completeness Uniform Both [HIGH]

For $U(\theta, \theta + 1)$, complete sufficient statistic is:

- A) $(X_{(1)}, X_{(n)})$
- B) $X_{(n)} - X_{(1)}$ (range)
- C) $\bar{X}$
- D) No complete sufficient statistic exists

Answer: A

Trap/Note: Range $X_{(n)} - X_{(1)}$ is ancillary (not sufficient); $(X_{(1)}, X_{(n)})$ is minimal sufficient and complete

Q91. Factorization Twoparam [HIGH]

For $N(\mu, \sigma^2)$ with both unknown, joint likelihood factors through:

- A) $(\bar{X}, \sum (X_i - \bar{X})^2)$
- B) $(\bar{X}, S^2)$ where S^2 uses $n - 1$
- C) $(\sum X_i, \sum X_i^2)$
- D) All of the above

Answer: D

Trap/Note: All are equivalent representations of the same sufficient statistic; factorization theorem doesn't specify unique form

Q92. Sufficient Not Complete [HIGH]

An example of sufficient but NOT complete statistic:

- A) $\bar{X}$ for $N(\mu, 1)$
- B) $(X_{(1)}, X_{(n)})$ for $N(\mu, 1)$
- C) $\sum X_i$ for $\text{Poisson}(\lambda)$
- D) $X_{(n)}$ for $U(0, \theta)$

Answer: B

Trap/Note: $(X_{(1)}, X_{(n)})$ is sufficient for μ in $N(\mu, 1)$ but NOT complete; $E[X_{(n)} + X_{(1)} - 2\mu] = 0$ but $X_{(n)} + X_{(1)} - 2\mu \neq 0$ a.s.

Q93. Basu Theorem Application [HIGH]

For $N(\mu, \sigma^2)$ with σ^2 known, by Basu's theorem:

- A) $\bar{X}$ and S^2 are independent
- B) $\bar{X}$ and sample range are independent
- C) S^2 and $\bar{X}$ are independent (since $\bar{X}$ is complete sufficient and S^2 is ancillary)
- D) All order statistics are independent of $\bar{X}$

Answer: B

Trap/Note: $\bar{X}$ is complete sufficient for μ (with σ^2 known); range is ancillary for μ ; by Basu, they're independent

Q94. Factorization Pareto [MEDIUM]

For $X_1, \dots, X_n \sim \text{Pareto}(\theta)$, $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, sufficient statistic is:

- A) $\sum X_i$
- B) $\prod X_i$
- C) $\sum \log X_i$
- D) $X_{(1)}$

Answer: C

Trap/Note: $\log f = n \log \theta - (\theta + 1) \sum \log x_i$; $T = \sum \log X_i$ is sufficient; $\prod X_i = \exp(\sum \log X_i)$ equivalent

Q95. Complete Stat Exponential Family [HIGH]

For k -parameter exponential family with natural parameters $\eta_1, \dots, \eta_k$ varying freely, the natural sufficient statistic $(T_1, \dots, T_k)$ is:

- A) Sufficient only
- B) Complete sufficient
- C) Minimal sufficient only
- D) Complete but not sufficient

Answer: B

Trap/Note: Full-rank exponential family: natural sufficient statistic is BOTH complete AND minimal sufficient; "full-rank" = natural parameter space has non-empty interior

Q96. Factorization Geometric [MEDIUM]

For $X_1, \dots, X_n \sim \text{Geometric}(p)$, sufficient statistic is:

- A) $X_{(1)}$
- B) $\sum X_i$
- C) $\bar{X}$
- D) Both B and C

Answer: D

Trap/Note: $\log f = \sum x_i \log(1-p) + n \log p$; $T = \sum X_i$ is sufficient; $\bar{X}$ is equivalent

Q97. Sufficiency Independence [HIGH]

If T is sufficient for θ and U is independent of T , then:

- A) U is ancillary
- B) (T, U) is sufficient for θ
- C) U contains no information about θ
- D) All of the above

Answer: D

Trap/Note: Independence from sufficient statistic implies ancillarity; (T, U) is trivially sufficient; ancillary statistics carry no θ -information

Q98. Minimal Sufficient Ratio [HIGH]

Lehmann-Scheffé method finds minimal sufficient statistic via:

- A) Ratio $f(\mathbf{x}; \theta)/f(\mathbf{y}; \theta)$ constant in θ iff $T(\mathbf{x}) = T(\mathbf{y})$
- B) Maximizing likelihood over θ
- C) Finding complete statistic
- D) Rao-Blackwell theorem

Answer: A

Trap/Note: The ratio method: T is minimal sufficient iff the ratio of likelihoods is free of θ exactly when $T(\mathbf{x}) = T(\mathbf{y})$

Q99. Complete Stat Discrete [HIGH]

For $X \sim \text{Bin}(n, p)$ with n known, X itself is complete sufficient because:

- A) It is the MLE
- B) $E_p[g(X)] = \sum_{x=0}^n g(x) \binom{n}{x} p^x (1-p)^{n-x} = 0$ for all p implies $g \equiv 0$
- C) It is a function of order statistics
- D) It attains CRLB

Answer: B

Trap/Note: Completeness proof uses the fact that a polynomial in $p/(1-p)$ that is identically zero must have all coefficients zero

Q100. Factorization Hypergeometric [HIGH]

For Hypergeometric(N, M, n) with M unknown, sufficient statistic is:

- A) $\bar{X}$
- B) $\sum X_i$ (number of successes in sample)
- C) $X_{(1)}$
- D) No reduction possible

Answer: B

Trap/Note: Hypergeometric is NOT exponential family but $\sum X_i$ is still sufficient; factorization applies directly

Q101. Sufficiency Transformation [MEDIUM]

If T is sufficient for θ and $\phi = g(T)$ is one-to-one, then:

- A) ϕ is also sufficient for θ
- B) ϕ is sufficient only if g is monotone
- C) ϕ is not necessarily sufficient
- D) ϕ is sufficient only if g is linear

Answer: A

Trap/Note: One-to-one functions of sufficient statistics are sufficient; the factorization still holds through the bijection

Q102. Complete Stat Location [HIGH]

For location family $f(x - \theta)$ with $\theta \in \mathbb{R}$, the order statistics $(X_{(1)}, \dots, X_{(n)})$ are:

- A) Complete sufficient
- B) Minimal sufficient but NOT complete
- C) Sufficient but not minimal
- D) Neither sufficient nor complete

Answer: B

Trap/Note: For location families, order statistics are minimal sufficient but NOT complete; spacings $X_{(i+1)} - X_{(i)}$ are ancillary, providing non-trivial functions with zero expectation

Q103. Factorization Multinomial [HIGH]

For Multinomial($n; p_1, \dots, p_k$) with $\sum p_i = 1$, sufficient statistic is:

- A) $(n_1, \dots, n_{k-1})$ (cell counts, last determined by constraint)
- B) $(n_1, \dots, n_k)$
- C) $n_1/n, \dots, n_{k-1}/n$ (proportions)
- D) All of the above

Answer: D

Trap/Note: All are equivalent representations; $(n_1, \dots, n_{k-1})$ determines $n_k = n - \sum_{i=1}^{k-1} n_i$

Q104. Sufficiency Nuisance [HIGH]

When $\theta = (\psi, \lambda)$ with λ nuisance, T is sufficient for ψ in the presence of λ if:

- A) T is sufficient for $\theta = (\psi, \lambda)$ jointly
- B) Conditional distribution of X given T is free of ψ (not necessarily λ)
- C) T is ancillary for λ
- D) T is complete for ψ

Answer: A

Trap/Note: Joint sufficiency implies marginal sufficiency; conditional sufficiency (given λ) is a weaker concept

Q105. Complete Stat Twosample [HIGH]

For two independent samples $X_i \sim N(\mu_1, \sigma^2)$ and $Y_j \sim N(\mu_2, \sigma^2)$ with common σ^2 , complete sufficient statistic is:

- A) $(\bar{X}, \bar{Y}, S_p^2)$ where S_p^2 is pooled variance
- B) $(\bar{X}, \bar{Y}, \sum (X_i - \bar{X})^2 + \sum (Y_j - \bar{Y})^2)$
- C) $(\sum X_i, \sum Y_j, \sum X_i^2 + \sum Y_j^2)$
- D) All of the above

Answer: D

Trap/Note: All are equivalent; three-dimensional sufficient statistic for three parameters (μ_1, μ_2, σ^2)

Q106. Factorization Inverse Gaussian [HIGH]

For $X \sim IG(\mu, \lambda)$ with μ known, sufficient statistic for λ is:

- A) $\sum X_i$
- B) $\sum (1/X_i - 1/\mu)$
- C) $\sum (X_i - \mu)^2/X_i$
- D) $\bar{X}$

Answer: C

Trap/Note: $\log f \propto \lambda \sum [(x_i - \mu)^2/x_i]$; $T = \sum (X_i - \mu)^2/X_i$ is sufficient for λ

Q107. Completeness Truncated Family [HIGH]

For Poisson(λ) truncated to $\{1, 2, 3, \dots\}$ (zero-truncated), $\sum X_i$ is:

- A) Complete sufficient
- B) Sufficient but not complete
- C) Complete but not sufficient
- D) Neither

Answer: A

Trap/Note: Zero-truncated Poisson is still exponential family (with modified normalizing constant); $\sum X_i$ remains complete sufficient

Q108. Sufficient Stat Regression [HIGH]

In simple linear regression $Y_i = \alpha + \beta x_i + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$, sufficient statistic for $(\alpha, \beta, \sigma^2)$ is:

- A) $(\hat{\alpha}, \hat{\beta}, \hat{\sigma}^2)$ — OLS estimates
- B) $(\sum Y_i, \sum x_i Y_i, \sum Y_i^2)$
- C) $(\bar{Y}, \sum (x_i - \bar{x}) Y_i, \text{RSS})$

- D) All of the above

Answer: D

Trap/Note: All are equivalent three-dimensional sufficient statistics; OLS estimates are one-to-one functions of the natural sufficient statistics

Q109. Basu Theorem Proof Idea [HIGH]

Basu's theorem proof relies on:

- A) Completeness of T and ancillarity of A to show $P(A \in B|T = t) = P(A \in B)$
- B) Independence of order statistics
- C) Factorization theorem
- D) Rao-Blackwell theorem

Answer: A

Trap/Note: Key step: $E_\theta[P(A \in B|T) - P_\theta(A \in B)] = 0$ for all θ ; by completeness, $P(A \in B|T) = P_\theta(A \in B)$ a.s., giving independence

Q110. Factorization Logistic Regression [HIGH]

In logistic regression with $P(Y_i = 1) = \frac{e^{\beta x_i}}{1 + e^{\beta x_i}}$, sufficient statistic for β is:

- A) $\sum Y_i$
- B) $\sum x_i Y_i$
- C) $(\sum Y_i, \sum x_i Y_i)$
- D) $\bar{Y}$

Answer: B

Trap/Note: $\log L = \beta \sum x_i y_i - \sum \log(1 + e^{\beta x_i})$; $T = \sum x_i Y_i$ is sufficient for β (single parameter)

Q111–Q160. Maximum Likelihood Estimation

Q111. MLE Normal Both [MEDIUM]

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, MLE of σ^2 is:

- A) $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$
- B) $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$
- C) $\frac{1}{n} \sum (X_i - \mu)^2$
- D) $\frac{1}{n+1} \sum (X_i - \bar{X})^2$

Answer: B

Trap/Note: MLE uses n denominator (biased); unbiased estimator uses $n - 1$; C is MLE only when μ is known

Q112. MLE Uniform [MEDIUM]

For $X_1, \dots, X_n \sim U(0, \theta)$, MLE of θ is:

- A) $2\bar{X}$
- B) $\bar{X}$
- C) $X_{(n)}$
- D) $\frac{n+1}{n} X_{(n)}$

Answer: C

Trap/Note: Likelihood $L(\theta) = \theta^{-n} I(\theta \geq X_{(n)})$ is decreasing in θ ; minimized at $\theta = X_{(n)}$; D is UMVUE not MLE

Q113. MLE Shifted Exponential [HIGH]

For $X_1, \dots, X_n \sim e^{-(x-\theta)}$, $x > \theta$, MLE of θ is:

- A) $\bar{X} - 1$
- B) $X_{(1)} - 1/n$
- C) $X_{(1)}$
- D) $\bar{X}$

Answer: C

Trap/Note: Likelihood $L(\theta) = e^{-\sum (x_i - \theta)} I(\theta \leq X_{(1)}) = e^{n\theta} e^{-\sum x_i} I(\theta \leq X_{(1)})$; increasing in θ ; maximized at $\theta = X_{(1)}$

Q114. MLE Cauchy [HIGH]

For $X_1, \dots, X_n \sim \text{Cauchy}(\theta, 1)$, MLE of θ :

- A) Is $\bar{X}$
- B) Is sample median
- C) Has no closed form; requires numerical optimization
- D) Is $X_{(n/2)}$ (middle order statistic)

Answer: C

Trap/Note: Score equation $\sum \frac{2(x_i - \theta)}{1 + (x_i - \theta)^2} = 0$ has no closed form; sample median is a good starting point but MLE $\neq$ median exactly

Q115. MLE Invariance Example [MEDIUM]

If $\hat{\lambda} = \bar{X}$ is MLE of λ for Poisson, MLE of $P(X = 0) = e^{-\lambda}$ is:

- A) $e^{-\bar{X}}$
- B) $\frac{1}{n} \sum I(X_i = 0)$
- C) $e^{-\lambda}$ evaluated at UMVUE
- D) $1 - \bar{X}$

Answer: A

Trap/Note: By invariance of MLE: $\widehat{e^{-\lambda}} = e^{-\hat{\lambda}} = e^{-\bar{X}}$; B is unbiased but not MLE; note $e^{-\bar{X}} \neq$ UMVUE

Q116. MLE Consistency [MEDIUM]

Under regularity conditions, MLE $\hat{\theta}_n$ satisfies:

- A) $\hat{\theta}_n \xrightarrow{p} \theta_0$ (true value)
- B) $\hat{\theta}_n \xrightarrow{a.s.} \theta_0$
- C) $\hat{\theta}_n$ is unbiased for all n
- D) Both A and B

Answer: A

Trap/Note: MLE is consistent (in probability) under regularity; strong consistency (a.s.) requires additional conditions; C is generally false

Q117. MLE Asymptotic Normality [MEDIUM]

Under regularity conditions, $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d}$:

- A) $N(0, I(\theta))$
- B) $N(0, 1/I(\theta))$
- C) $N(0, I(\theta)^{-1})$
- D) Both B and C (same thing for scalar θ)

Answer: D

Trap/Note: For scalar θ : $1/I(\theta)$ and $I(\theta)^{-1}$ are identical; for vector θ : $I(\theta)^{-1}$ is the matrix inverse

Q118. MLE Multimodal [HIGH]

When log-likelihood has multiple local maxima, MLE is:

- A) The global maximum
- B) The local maximum closest to MOM estimate
- C) The average of all local maxima
- D) Undefined

Answer: A

Trap/Note: MLE is defined as the GLOBAL maximum; in practice, numerical methods may find local maxima — this is a computational challenge, not a definitional one

Q119. MLE Boundary [HIGH]

For $X_1, \dots, X_n \sim N(\mu, 1)$ with constraint $\mu \geq 0$, MLE of μ is:

- A) $\bar{X}$ always
- B) $\max(\bar{X}, 0)$
- C) $|\bar{X}|$
- D) $\bar{X}$ if $\bar{X} > 0$, undefined otherwise

Answer: B

Trap/Note: Constrained MLE: if unconstrained MLE $\bar{X} \geq 0$, use it; if $\bar{X} < 0$, boundary value $\mu = 0$ maximizes constrained likelihood

Q120. MLE Discrete Uniform [HIGH]

For $X_1, \dots, X_n \sim \text{DiscreteUniform}\{1, 2, \dots, N\}$ with N unknown, MLE of N is:

- A) $2\bar{X} - 1$
- B) $X_{(n)}$
- C) $X_{(n)} + 1$
- D) $\lceil 2\bar{X} \rceil$

Answer: B

Trap/Note: $L(N) = N^{-n}I(N \geq X_{(n)})$; decreasing in N ; MLE = $X_{(n)}$ (smallest valid N); note MLE is biased

Q121. MLE Mixture EM [HIGH]

EM algorithm for MLE in mixture models:

- A) E-step: compute expected complete-data log-likelihood; M-step: maximize it
- B) E-step: maximize; M-step: compute expectation
- C) Directly maximizes observed-data likelihood
- D) Requires complete data

Answer: A

Trap/Note: EM: E-step computes $Q(\theta|\theta^{(t)}) = E[\log L_c | \text{observed}, \theta^{(t)}]$; M-step maximizes Q ; guaranteed to increase observed likelihood

Q122. MLE Exponential Family [MEDIUM]

In exponential family, MLE satisfies:

- A) $E_{\hat{\theta}}[T(X)] = \bar{T}$ (observed sufficient statistic equals expected)
- B) Score equation: $\sum \partial \log f / \partial \theta = 0$
- C) Both A and B
- D) Neither; MLE requires numerical methods

Answer: C

Trap/Note: In exponential family, score equation reduces to moment equation $E_\theta[T] = \bar{T}$; both characterizations are equivalent

Q123. MLE Regularity Failure [HIGH]

MLE may fail to be consistent when:

- **A)** Number of parameters grows with sample size (Neyman-Scott problem)
- **B)** Distribution is normal
- **C)** Sample size is large
- **D)** Likelihood is unimodal

Answer: A

Trap/Note: Neyman-Scott: incidental parameters problem; MLE of variance in panel data with individual fixed effects is inconsistent

Q124. MLE Poisson Truncated [HIGH]

For zero-truncated Poisson, MLE $\hat{\lambda}$ satisfies:

- **A)** $\hat{\lambda} = \bar{X}$
- **B)** $\bar{X} = \hat{\lambda}/(1 - e^{-\hat{\lambda}})$
- **C)** $\hat{\lambda} = \bar{X} - 1$
- **D)** $\hat{\lambda} = \bar{X}(1 - e^{-\bar{X}})$

Answer: B

Trap/Note: Truncated mean $E[X|X > 0] = \lambda/(1 - e^{-\lambda})$; setting equal to $\bar{X}$ gives implicit equation; no closed form

Q125. MLE Gamma Shape [HIGH]

For $\Gamma(\alpha, \beta)$ with both unknown, MLE of α satisfies:

- **A)** $\log \hat{\alpha} - \psi(\hat{\alpha}) = \log \bar{X} - \overline{\log X}$
- **B)** $\hat{\alpha} = \bar{X}/\hat{\beta}$
- **C)** $\hat{\alpha} = n/\sum \log X_i$
- **D)** No closed form exists; use Newton-Raphson

Answer: A

Trap/Note: ψ is digamma function; A is the score equation for α ; D is correct in spirit but A specifies the equation; B gives $\hat{\alpha}$ in terms of $\hat{\beta}$

Q126. MLE Logistic [HIGH]

For logistic regression, MLE:

- **A)** Has closed form via OLS
- **B)** Requires iterative methods (IRLS or Newton-Raphson)
- **C)** Is always unique

- D) Both B and C

Answer: D

Trap/Note: Logistic MLE: no closed form; IRLS converges to unique global maximum (log-likelihood is concave); uniqueness requires full rank design matrix

Q127. MLE Censored Exponential [HIGH]

For Type-II censored sample (smallest r out of n observed) from $\text{Exp}(\theta)$, MLE of θ is:

- A) $1/\bar{X}_r$ where $\bar{X}_r$ is mean of r observed
- B) $r/[\sum_{i=1}^r X_{(i)} + (n-r)X_{(r)}]$
- C) $r/\sum_{i=1}^r X_{(i)}$
- D) $1/X_{(r)}$

Answer: B

Trap/Note: Total time on test = $\sum_{i=1}^r X_{(i)} + (n-r)X_{(r)}$; MLE = $r/\text{total time on test}$; C ignores censored contribution

Q128. MLE Weibull Shape [HIGH]

For Weibull(k, λ) with λ known, MLE of k satisfies:

- A) $1/\hat{k} + \overline{\log X} - \frac{\sum X_i^{\hat{k}} \log X_i}{\sum X_i^{\hat{k}}} = 0$
- B) $\hat{k} = n / \sum \log X_i$
- C) $\hat{k} = \bar{X}/\lambda$
- D) $\hat{k} = 1/\overline{\log X}$

Answer: A

Trap/Note: Score equation for Weibull shape is transcendental; B is MLE for Beta($\theta, 1$) not Weibull; requires numerical solution

Q129. MLE Dirichlet [HIGH]

For Dirichlet($\alpha_1, \dots, \alpha_k$), MLE of α_j satisfies:

- A) $\psi(\hat{\alpha}_j) - \psi(\hat{\alpha}_0) = \overline{\log X_j}$ where $\alpha_0 = \sum \alpha_j$
- B) $\hat{\alpha}_j = \bar{X}_j$
- C) $\hat{\alpha}_j = n\bar{X}_j$
- D) No MLE exists

Answer: A

Trap/Note: ψ is digamma; system of equations must be solved simultaneously; B/C are MOM not MLE

Q130. MLE Nonregular Rate [HIGH]

For $X_1, \dots, X_n \sim U(\theta - 1/2, \theta + 1/2)$, MLE of θ :

- A) Is $\bar{X}$
- B) Is any value in $[X_{(n)} - 1/2, X_{(1)} + 1/2]$
- C) Is $\frac{X_{(1)} + X_{(n)}}{2}$
- D) Does not exist

Answer: B

Trap/Note: Likelihood is constant for $\theta \in [X_{(n)} - 1/2, X_{(1)} + 1/2]$; MLE is NOT unique; $\frac{X_{(1)} + X_{(n)}}{2}$ is a natural choice but not the unique MLE

Q131. MLE Convergence Rate [HIGH]

Under regularity, MLE converges at rate:

- A) $n^{-1/2}$ (parametric rate)
- B) n^{-1}
- C) $n^{-1/3}$
- D) $(\log n)^{-1}$

Answer: A

Trap/Note: Parametric MLE: $\hat{\theta} - \theta = O_p(n^{-1/2})$; nonparametric rates are slower; $n^{-1/2}$ is the hallmark of parametric efficiency

Q132. MLE Profile Likelihood [HIGH]

Profile likelihood for ψ (with nuisance λ) is:

- A) $L_p(\psi) = \max_{\lambda} L(\psi, \lambda)$
- B) $L_p(\psi) = \int L(\psi, \lambda) d\lambda$
- C) $L_p(\psi) = L(\psi, \hat{\lambda})$ where $\hat{\lambda}$ is marginal MLE
- D) $L_p(\psi) = L(\psi, \hat{\lambda}_{\psi})$ where $\hat{\lambda}_{\psi} = \arg \max_{\lambda} L(\psi, \lambda)$

Answer: D

Trap/Note: Profile likelihood: maximize over nuisance at each fixed ψ ; A and D are equivalent; B is marginal likelihood (Bayesian); C uses wrong $\hat{\lambda}$

Q133. MLE Second Order [HIGH]

Bartlett correction improves likelihood ratio test by:

- A) Scaling $-2 \log \Lambda$ by factor $1 + b/n$ to achieve χ^2 distribution to order $O(n^{-2})$
- B) Adding b/n to the test statistic
- C) Using F -distribution instead of χ^2
- D) Adjusting degrees of freedom

Answer: A

Trap/Note: Bartlett correction: divide $-2 \log \Lambda$ by $(1 + b/n)$; improves χ^2 approximation from $O(n^{-1})$ to $O(n^{-2})$ error

Q134. MLE Penalized [HIGH]

Penalized MLE (e.g., LASSO) maximizes:

- **A)** $\log L(\theta) - \lambda \|\theta\|_1$
- **B)** $\log L(\theta) + \lambda \|\theta\|_1$
- **C)** $\log L(\theta) - \lambda \|\theta\|_2^2$
- **D)** A only (LASSO uses L_1 penalty)

Answer: A

Trap/Note: LASSO: L_1 penalty subtracted; Ridge: L_2^2 penalty subtracted; both are A-type (subtract penalty); D is correct but incomplete

Q135. MLE Robust [HIGH]

M-estimators generalize MLE by solving:

- **A)** $\sum \psi(X_i; \theta) = 0$ for some function ψ
- **B)** $\sum \partial \log f / \partial \theta = 0$ only
- **C)** $\min \sum \rho(X_i; \theta)$ for loss ρ
- **D)** Both A and C (equivalent formulations)

Answer: D

Trap/Note: M-estimators: $\psi = \partial \rho / \partial \theta$; MLE is special case with $\rho = -\log f$; Huber M-estimator uses bounded ψ

Q136. MLE Exponential Rate [MEDIUM]

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$ (rate parametrization), MLE of λ is:

- **A)** $\bar{X}$
- **B)** $1/\bar{X}$
- **C)** $n / \sum X_i$
- **D)** Both B and C

Answer: D

Trap/Note: $1/\bar{X} = n / \sum X_i$ — identical; both are correct expressions for MLE of rate λ

Q137. MLE Beta Both [HIGH]

For $X \sim \text{Beta}(\alpha, \beta)$ with both unknown, MLE satisfies:

- **A)** $\psi(\hat{\alpha}) - \psi(\hat{\alpha} + \hat{\beta}) = \overline{\log X}$ and $\psi(\hat{\beta}) - \psi(\hat{\alpha} + \hat{\beta}) = \overline{\log(1 - X)}$
- **B)** $\hat{\alpha} = \bar{X}$, $\hat{\beta} = 1 - \bar{X}$
- **C)** $\hat{\alpha} / (\hat{\alpha} + \hat{\beta}) = \bar{X}$
- **D)** No MLE exists for Beta with both parameters unknown

Answer: A

Trap/Note: System of two digamma equations; C gives one equation (moment condition) but not MLE; requires numerical solution

Q138. MLE Nonidentifiable [HIGH]

MLE fails when model is non-identifiable because:

- A) Multiple parameter values give same likelihood; MLE is not unique
- B) Likelihood is unbounded
- C) Score equation has no solution
- D) Fisher information is zero

Answer: A

Trap/Note: Non-identifiability: $L(\theta_1) = L(\theta_2)$ for $\theta_1 \neq \theta_2$; MLE set is not a singleton; Fisher information matrix is singular

Q139. MLE Separated Data [HIGH]

In logistic regression, MLE does not exist when:

- A) Sample size is small
- B) Data is perfectly separated (one predictor perfectly predicts outcome)
- C) Predictors are correlated
- D) Response is binary

Answer: B

Trap/Note: Perfect separation: likelihood is monotone increasing; $\text{MLE} \rightarrow \pm\infty$; Firth's penalized likelihood is a remedy

Q140. MLE Consistency Proof [HIGH]

Wald's proof of MLE consistency uses:

- A) Law of large numbers applied to log-likelihood ratio
- B) Central limit theorem
- C) Rao-Blackwell theorem
- D) Factorization theorem

Answer: A

Trap/Note: Key: $\frac{1}{n} \log \frac{L(\theta)}{L(\theta_0)} \xrightarrow{p} E_{\theta_0}[\log \frac{f(X;\theta)}{f(X;\theta_0)}] \leq 0$ by Jensen (Kullback-Leibler divergence ≥ 0)

Q141. MLE Poisson Regression [MEDIUM]

In Poisson regression with $\log \mu_i = \mathbf{x}_i^T \beta$, score equation for β is:

- A) $\sum \mathbf{x}_i(Y_i - \mu_i) = \mathbf{0}$
- B) $\sum \mathbf{x}_i Y_i = \sum \mathbf{x}_i \mu_i$
- C) $X^T(Y - \mu) = \mathbf{0}$
- D) All of the above

Answer: D

Trap/Note: All three are equivalent forms of the Poisson GLM score equation; IRLS solves this iteratively

Q142. MLE Negative Binomial [HIGH]

For $\text{NegBin}(r, p)$ with r known, MLE of p is:

- A) $r/(r + \bar{X})$
- B) $\bar{X}/(r + \bar{X})$
- C) $r/\bar{X}$
- D) $1/\bar{X}$

Answer: A

Trap/Note: $E[X] = r(1 - p)/p$; solving $\bar{X} = r(1 - \hat{p})/\hat{p}$ gives $\hat{p} = r/(r + \bar{X})$; B is MLE of $1 - p$

Q143. MLE Zero Inflated [HIGH]

Zero-inflated Poisson (ZIP) has $P(X = 0) = \pi + (1 - \pi)e^{-\lambda}$. MLE requires:

- A) EM algorithm (latent class indicator)
- B) Direct score equations with closed form
- C) Method of moments only
- D) Bayesian methods

Answer: A

Trap/Note: ZIP is a mixture model; EM treats class membership as latent; E-step computes posterior probability of structural zero; M-step updates π and λ

Q144. MLE Survival Kaplan Meier [HIGH]

Kaplan-Meier estimator is:

- A) MLE of survival function under right censoring (nonparametric)
- B) MLE of exponential survival function
- C) Method of moments estimator
- D) Bayesian estimator

Answer: A

Trap/Note: KM is the nonparametric MLE of $S(t)$; it maximizes the nonparametric likelihood over all step functions; not restricted to parametric family

Q145. MLE Information Matrix Equality [HIGH]

Fisher information matrix equality states:

- A) $-E[\partial^2 \log L / \partial \theta \partial \theta^T] = E[(\partial \log L / \partial \theta)(\partial \log L / \partial \theta)^T]$
- B) $E[\partial \log L / \partial \theta] = 0$
- C) Both A and B
- D) $I(\theta) = \text{Var}(\partial \log L / \partial \theta)$

Answer: C

Trap/Note: B is the score equation property; A is the information matrix equality; D follows from B (since $E[\text{score}] = 0$, variance = second moment); all three are related

Q146. MLE Restricted Parameter [HIGH]

For $X \sim N(\theta, 1)$ with $\theta \in \{0, 1\}$ (discrete parameter space), MLE is:

- A) $\bar{X}$ rounded to nearest integer
- B) 0 if $\bar{X} < 1/2$, 1 if $\bar{X} \geq 1/2$
- C) $\bar{X}$ always
- D) 0 always (conservative)

Answer: B

Trap/Note: Compare $L(0) = e^{-n\bar{X}^2/2}$ vs $L(1) = e^{-n(\bar{X}-1)^2/2}$; $L(1) > L(0)$ iff $\bar{X} > 1/2$

Q147. MLE Curved Exponential Normal [HIGH]

For $X_1, \dots, X_n \sim N(\theta, \theta^2)$ (curved exponential family), MLE of θ is:

- A) $\bar{X}$
- B) S (sample SD)
- C) Solution to $\hat{\theta}^2 + \hat{\theta}\bar{X} - (\bar{X}^2 + S^2) = 0$ approximately
- D) $\sqrt{\bar{X}^2 + S^2}$

Answer: C

Trap/Note: Score equations for $\mu = \theta$ and $\sigma^2 = \theta^2$ must be combined; leads to quadratic in $\hat{\theta}$; no simple closed form

Q148. MLE Poisson UMVUE Comparison [HIGH]

For $\text{Poisson}(\lambda)$, MLE of $e^{-\lambda}$ is $e^{-\bar{X}}$. UMVUE is:

- A) $e^{-\bar{X}}$
- B) $(1 - 1/n)^{\sum X_i}$
- C) $e^{-\bar{X}}(1 + 1/(2n))$ (bias-corrected)
- D) $\frac{1}{n} \sum I(X_i = 0)$

Answer: B

Trap/Note: UMVUE of $e^{-\lambda} = P(X = 0)$: use Rao-Blackwell with $T = \sum X_i$; $E[I(X_1 = 0)|T = t] = (1 - 1/n)^t$; so UMVUE = $(1 - 1/n)^T$

Q149. MLE Asymptotic Variance [MEDIUM]

Asymptotic variance of MLE $\hat{\theta}$ is:

- A) $1/(nI(\theta))$
- B) $I(\theta)/n$
- C) $nI(\theta)$
- D) $1/I(\theta)$

Answer: A

Trap/Note: $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1/I(\theta))$; so $\text{AsyVar}(\hat{\theta}) = 1/(nI(\theta))$; D is for single observation

Q150. MLE Score Test [HIGH]

Rao's score test statistic is:

- A) $\frac{[\partial \log L / \partial \theta]_{\theta=\theta_0}]^2}{nI(\theta_0)} \xrightarrow{d} \chi_1^2$ under H_0
- B) $-2 \log \Lambda \xrightarrow{d} \chi_1^2$
- C) $\frac{(\hat{\theta} - \theta_0)^2}{1/(nI(\hat{\theta}))} \xrightarrow{d} \chi_1^2$
- D) All equivalent asymptotically

Answer: D

Trap/Note: Wald (C), Score/Rao (A), and LRT (B) are asymptotically equivalent under H_0 ; they differ in finite samples and under alternatives

Q151. MLE Exponential Mean [MEDIUM]

For $X_1, \dots, X_n \sim \text{Exp}(\mu)$ (mean parametrization), MLE of μ is:

- A) $1/\bar{X}$
- B) $\bar{X}$
- C) $n/\sum X_i$
- D) $\bar{X}$ (same as B)

Answer: B

Trap/Note: Mean parametrization: $f(x; \mu) = \frac{1}{\mu} e^{-x/\mu}$; MLE = $\bar{X}$; rate parametrization: MLE = $1/\bar{X}$; know which parametrization is used

Q152. MLE Multivariate Normal [HIGH]

For $X_1, \dots, X_n \sim N_p(\mu, \Sigma)$ with both unknown, MLE of Σ is:

- A) $\frac{1}{n} \sum (X_i - \bar{X})(X_i - \bar{X})^T$
- B) $\frac{1}{n-1} \sum (X_i - \bar{X})(X_i - \bar{X})^T$
- C) $\frac{1}{n-p} \sum (X_i - \bar{X})(X_i - \bar{X})^T$
- D) Sample covariance matrix

Answer: A

Trap/Note: MLE uses n denominator (biased); unbiased estimator uses $n-1$; $n-p$ appears in regression residual variance

Q153. MLE Dirichlet Multinomial [HIGH]

For Dirichlet-Multinomial model, MLE of Dirichlet parameters:

- A) Has closed form via method of moments
- B) Requires solving system of digamma equations iteratively
- C) Is simply cell proportions
- D) Does not exist

Answer: B

Trap/Note: Same structure as Dirichlet MLE; digamma equations; Minka's fixed-point iteration is commonly used

Q154. MLE Nonparametric [HIGH]

Nonparametric MLE of CDF F based on i.i.d. sample is:

- A) Normal CDF fitted to data
- B) Empirical CDF $\hat{F}_n(x) = \frac{1}{n} \sum I(X_i \leq x)$
- C) Kernel density estimate
- D) Smoothed empirical CDF

Answer: B

Trap/Note: NPMLE of CDF is the empirical CDF (Glivenko-Cantelli); it maximizes the non-parametric likelihood $\prod f(X_i)$ over all distributions

Q155. MLE Exponential Family Canonical [HIGH]

In canonical exponential family, MLE $\hat{\eta}$ of natural parameter satisfies:

- A) $\nabla A(\hat{\eta}) = \bar{T}$ where A is log-partition function
- B) $\hat{\eta} = \bar{T}$
- C) $\hat{\eta}$ minimizes KL divergence from empirical distribution
- D) Both A and C

Answer: D

Trap/Note: $\nabla A(\eta) = E_\eta[T(X)]$; setting equal to $\bar{T}$ gives MLE; this is equivalent to minimizing KL divergence (maximum entropy principle)

Q156. MLE Robust Breakdown [HIGH]

Breakdown point of MLE for normal location parameter is:

- A) 0 (single outlier can make MLE arbitrarily large)
- B) 1/2 (50% contamination)
- C) 1/n
- D) 1/4

Answer: A

Trap/Note: Sample mean (MLE for normal location) has breakdown point 0; one extreme outlier destroys it; median has breakdown point 1/2

Q157. MLE Conditional [HIGH]

Conditional MLE (conditioning on sufficient statistic for nuisance) is used to:

- A) Eliminate nuisance parameters from likelihood
- B) Increase efficiency
- C) Reduce computation
- D) Handle missing data

Answer: A

Trap/Note: Conditional likelihood eliminates nuisance parameters; used in matched case-control studies, logistic regression with fixed effects

Q158. MLE Marginal [HIGH]

Marginal MLE (integrating out random effects) is used in:

- A) Fixed effects models
- B) Mixed effects models (REML is a variant)
- C) Nonparametric models
- D) Saturated models

Answer: B

Trap/Note: REML (Restricted MLE) maximizes marginal likelihood after integrating out fixed effects; used for variance component estimation in mixed models

Q159. MLE Semiparametric [HIGH]

In Cox proportional hazards model, partial likelihood MLE:

- A) Estimates baseline hazard and regression coefficients jointly
- B) Estimates regression coefficients without specifying baseline hazard
- C) Requires parametric baseline hazard
- D) Is equivalent to full MLE

Answer: B

Trap/Note: Cox partial likelihood: baseline hazard cancels; only regression coefficients estimated; semiparametric efficiency theory applies

Q160. MLE Information Observed [HIGH]

Observed Fisher information $\hat{I}(\hat{\theta}) = -\partial^2 \log L / \partial \theta^2|_{\hat{\theta}}$ vs expected $I(\theta)$:

- A) Observed is always larger
- B) Expected is always larger
- C) They are asymptotically equivalent; observed often preferred in practice
- D) They are always equal

Answer: C

Trap/Note: Both are consistent for $I(\theta_0)$; Efron-Hinkley showed observed information is preferred for conditional inference; neither dominates in finite samples

Q161–Q200. Rao–Blackwell, UMVUE & Method of Moments

Q161. Rao Blackwell Statement [MEDIUM]

Rao-Blackwell theorem states: if T_0 is unbiased for $g(\theta)$ and S is sufficient, then $T^* = E[T_0|S]$:

- A) Has smaller or equal variance than T_0
- B) Is unbiased for $g(\theta)$
- C) Is a function of S
- D) All of the above

Answer: D

Trap/Note: All three properties hold: unbiasedness (tower property), variance reduction (conditional variance formula), and T^* is a function of S by definition

Q162. Rao Blackwell Example Bernoulli [MEDIUM]

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, improve $T_0 = X_1$ (unbiased for p) using $S = \sum X_i$:

- A) $E[X_1 | \sum X_i = t] = t/n$
- B) $E[X_1 | \sum X_i = t] = t(t-1)/[n(n-1)]$
- C) $E[X_1 | \sum X_i = t] = t^2/n^2$
- D) $E[X_1 | \sum X_i = t] = 1/n$

Answer: A

Trap/Note: $P(X_1 = 1 | \sum X_i = t) = \binom{n-1}{t-1} / \binom{n}{t} = t/n$; result is $\bar{X}$ — the UMVUE

Q163. UMVUE Poisson Zero [HIGH]

For $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$, UMVUE of $P(X = 0) = e^{-\lambda}$ is:

- A) $e^{-\bar{X}}$
- B) $(1 - 1/n)^{\sum X_i}$
- C) $\frac{1}{n} \sum I(X_i = 0)$
- D) $e^{-\sum X_i/n}$

Answer: B

Trap/Note: $E[I(X_1 = 0) | T = t]$ where $T = \sum X_i \sim \text{Poisson}(n\lambda)$: $P(X_1 = 0 | T = t) = (1 - 1/n)^t$; so UMVUE = $(1 - 1/n)^T$

Q164. UMVUE Normal Sigma [HIGH]

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with μ known, UMVUE of σ^2 is:

- A) $\frac{1}{n} \sum (X_i - \mu)^2$
- B) $\frac{1}{n-1} \sum (X_i - \mu)^2$
- C) $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$
- D) $\frac{1}{n} \sum (X_i - \bar{X})^2$

Answer: A

Trap/Note: When μ is KNOWN, $\sum (X_i - \mu)^2/n$ is UMVUE (attains CRLB); when μ unknown, UMVUE is S^2 with $n - 1$ denominator

Q165. UMVUE Uniform Theta Sq [HIGH]

For $X_1, \dots, X_n \sim U(0, \theta)$, UMVUE of θ^2 is:

- A) $X_{(n)}^2$
- B) $\frac{(n+1)(n+2)}{n^2} X_{(n)}^2$
- C) $\frac{n+2}{n} X_{(n)}^2$
- D) $4\bar{X}^2/3$

Answer: B

Trap/Note: $E[X_{(n)}^k] = \frac{n\theta^k}{n+k}$; for $k=2$: $E[X_{(n)}^2] = \frac{n\theta^2}{n+2}$; UMVUE = $\frac{(n+1)(n+2)}{n^2} X_{(n)}^2$

Q166. UMVUE Exponential Function [HIGH]

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$, UMVUE of λ^k is:

- A) $(1/\bar{X})^k$
- B) $\frac{n!}{(n-k)!n^k} \cdot \frac{1}{\bar{X}^k}$ (for $k < n$)
- C) $\bar{X}^{-k}$
- D) $\frac{\Gamma(n)}{\Gamma(n-k)} \cdot \frac{1}{(\sum X_i)^k}$

Answer: D

Trap/Note: $T = \sum X_i \sim \Gamma(n, 1/\lambda)$; $E[T^{-k}] = \frac{\lambda^k \Gamma(n-k)}{\Gamma(n)}$; UMVUE of $\lambda^k = \frac{\Gamma(n)}{\Gamma(n-k)} T^{-k}$

Q167. UMVUE Bernoulli P Sq [HIGH]

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, UMVUE of p^2 is:

- A) $\bar{X}^2$
- B) $\frac{T(T-1)}{n(n-1)}$ where $T = \sum X_i$
- C) T^2/n^2
- D) $\bar{X}(\bar{X} - 1/n)$

Answer: B

Trap/Note: $E[X_1 X_2 | T = t] = \frac{t(t-1)}{n(n-1)}$ (hypergeometric argument); UMVUE = $T(T-1)/[n(n-1)]$; note $\bar{X}^2$ is biased for p^2

Q168. UMVUE Poisson Lambda Sq [HIGH]

For $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$, UMVUE of λ^2 is:

- A) $\bar{X}^2$
- B) $\bar{X}^2 - \bar{X}/n$
- C) $T(T-1)/n^2$ where $T = \sum X_i$
- D) T^2/n^2

Answer: C

Trap/Note: $E[T(T-1)] = E[T^2] - E[T] = n^2\lambda^2 + n\lambda - n\lambda = n^2\lambda^2$; so UMVUE = $T(T-1)/n^2 = \bar{X}^2 - \bar{X}/n$; B and C are equivalent

Q169. Method Of Moments Gamma [MEDIUM]

For $X \sim \Gamma(\alpha, \beta)$, MOM estimators using first two moments:

- A) $\hat{\alpha} = \bar{X}^2/S^2, \hat{\beta} = S^2/\bar{X}$
- B) $\hat{\alpha} = \bar{X}/S^2, \hat{\beta} = S^2$
- C) $\hat{\alpha} = S^2/\bar{X}^2, \hat{\beta} = \bar{X}/S^2$
- D) $\hat{\alpha} = \bar{X}, \hat{\beta} = S^2/\bar{X}$

Answer: A

Trap/Note: $E[X] = \alpha\beta, \text{Var}(X) = \alpha\beta^2; \beta = \text{Var}/E = S^2/\bar{X}; \alpha = E/\beta = \bar{X}^2/S^2$

Q170. Method Of Moments Beta [MEDIUM]

For $X \sim \text{Beta}(\alpha, \beta)$, MOM estimators:

- A) $\hat{\alpha} = \bar{X}[\bar{X}(1 - \bar{X})/S^2 - 1], \hat{\beta} = (1 - \bar{X})[\bar{X}(1 - \bar{X})/S^2 - 1]$
- B) $\hat{\alpha} = \bar{X}, \hat{\beta} = 1 - \bar{X}$
- C) $\hat{\alpha} = \bar{X}/S^2, \hat{\beta} = (1 - \bar{X})/S^2$
- D) $\hat{\alpha} = \bar{X}^2/S^2, \hat{\beta} = (1 - \bar{X})^2/S^2$

Answer: A

Trap/Note: $E[X] = \alpha/(\alpha + \beta), \text{Var}(X) = \alpha\beta/[(\alpha + \beta)^2(\alpha + \beta + 1)]$; let $m = \bar{X}, v = S^2$;
 $\hat{\alpha} = m[m(1 - m)/v - 1]$

Q171. Rao Blackwell Improvement [HIGH]

Rao-Blackwell improvement is strict (strict inequality in variance) unless:

- A) T_0 is already a function of S
- B) T_0 is the MLE
- C) T_0 is the UMVUE
- D) Both A and C (equivalent)

Answer: D

Trap/Note: Variance reduction is strict unless T_0 is already a function of S ; if $T_0 = h(S)$, then $E[T_0|S] = T_0$ and no improvement occurs

Q172. UMVUE Existence [HIGH]

UMVUE exists if and only if:

- A) A complete sufficient statistic exists
- B) An unbiased estimator exists
- C) Both A and B
- D) The distribution is from exponential family

Answer: C

Trap/Note: Need both: (1) an unbiased estimator must exist, and (2) a complete sufficient statistic must exist; then Lehmann-Scheffé gives unique UMVUE

Q173. MOM Consistency [MEDIUM]

Method of moments estimators are consistent because:

- A) They are always unbiased
- B) Sample moments converge to population moments by LLN
- C) They attain CRLB
- D) They are functions of sufficient statistics

Answer: B

Trap/Note: LLN: $\bar{X}^k \xrightarrow{p} E[X^k]$; continuous mapping theorem: MOM estimators (continuous functions of sample moments) are consistent

Q174. UMVUE Normal Absolute Deviation [HIGH]

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, UMVUE of $|\mu|$:

- A) $|\bar{X}|$
- B) $|\bar{X}| \Phi(|\bar{X}| \sqrt{n}/\sigma) - \sigma \phi(|\bar{X}| \sqrt{n}/\sigma) / \sqrt{n}$ (complex expression)
- C) Does not exist (no unbiased estimator of $|\mu|$)
- D) $|\bar{X}|$ corrected for bias

Answer: B

Trap/Note: UMVUE of $|\mu|$ exists but has complex form involving normal CDF and PDF; $|\bar{X}|$ is biased for $|\mu|$; this is a classic hard ISS question

Q175. UMVUE Shifted Exponential [HIGH]

For $X_1, \dots, X_n \sim e^{-(x-\theta)}$, $x > \theta$, UMVUE of θ is:

- A) $X_{(1)}$
- B) $X_{(1)} - 1/n$
- C) $\bar{X} - 1$
- D) $X_{(1)} + 1/n$

Answer: B

Trap/Note: $E[X_{(1)}] = \theta + 1/n$; unbiased: $X_{(1)} - 1/n$; $X_{(1)}$ is complete sufficient; so $X_{(1)} - 1/n$ is UMVUE

Q176. MOM Negative Binomial [HIGH]

For $X \sim \text{NegBin}(r, p)$ with r known, MOM estimator of p :

- A) $r/(r + \bar{X})$
- B) $\bar{X}/(r + \bar{X})$
- C) $r/\bar{X}$
- D) $1/(1 + \bar{X}/r)$

Answer: A

Trap/Note: $E[X] = r(1-p)/p$; solving: $\hat{p} = r/(r + \bar{X})$; same as MLE in this case; D is equivalent to A

Q177. UMVUE Cauchy [HIGH]

For $X_1, \dots, X_n \sim \text{Cauchy}(\theta, 1)$, UMVUE of θ :

- A) Is the sample median
- B) Is $\bar{X}$
- C) Does not exist (no complete sufficient statistic)
- D) Is the MLE

Answer: C

Trap/Note: Cauchy is not exponential family; minimal sufficient statistic is all order statistics; no complete sufficient statistic exists; hence no UMVUE

Q178. Rao Blackwell Poisson Indicator [HIGH]

For $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$, improve $T_0 = I(X_1 = 1)$ using $T = \sum X_i$:

- A) $E[I(X_1 = 1)|T = t] = t(1 - 1/n)^{t-1}/n$
- B) $E[I(X_1 = 1)|T = t] = t/n$
- C) $E[I(X_1 = 1)|T = t] = e^{-1/n}(1/n)^t/t!$ (Poisson conditional)
- D) $E[I(X_1 = 1)|T = t] = \binom{t}{1}(1/n)^1(1 - 1/n)^{t-1}$

Answer: D

Trap/Note: Given $T = t$, $X_1 \sim \text{Bin}(t, 1/n)$; $P(X_1 = 1|T = t) = \binom{t}{1}(1/n)(1 - 1/n)^{t-1} = t(1 - 1/n)^{t-1}/n$; A and D are equivalent

Q179. UMVUE Uniform Probability [HIGH]

For $X_1, \dots, X_n \sim U(0, \theta)$, UMVUE of $P(X \leq c) = c/\theta$ (for fixed $c < \theta$) is:

- A) $c/X_{(n)}$
- B) $\min(c/X_{(n)}, 1)$
- C) $cn/[(n+1)X_{(n)}]$
- D) $\frac{n}{n+1} \cdot \frac{c}{X_{(n)}}$ if $X_{(n)} > c$, else 1

Answer: D

Trap/Note: Need to handle case $X_{(n)} \leq c$ (then $P(X \leq c) = 1$ certainly); UMVUE must be bounded in $[0, 1]$; careful case analysis required

Q180. MOM Lognormal [MEDIUM]

For $X \sim \text{LogNormal}(\mu, \sigma^2)$, MOM estimators using $E[X] = e^{\mu + \sigma^2/2}$ and $\text{Var}(X) = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$:

- A) $\hat{\sigma}^2 = \log(1 + S^2/\bar{X}^2)$, $\hat{\mu} = \log \bar{X} - \hat{\sigma}^2/2$
- B) $\hat{\mu} = \log \bar{X}$, $\hat{\sigma}^2 = S_{\log}^2$
- C) $\hat{\mu} = \log \bar{X}$, $\hat{\sigma}^2 = S^2/\bar{X}^2$
- D) Both A and B (A is MOM, B is MLE)

Answer: D

Trap/Note: MOM uses raw moments (A); MLE uses log-transformed data (B); they differ; B is also the UMVUE approach

Q181. UMVUE Bernoulli Variance [HIGH]

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, UMVUE of $p(1-p) = \text{Var}(X)$ is:

- A) $\bar{X}(1-\bar{X})$
- B) $\frac{T(n-T)}{n(n-1)}$ where $T = \sum X_i$
- C) $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$
- D) Both B and C (equivalent)

Answer: D

Trap/Note: $S^2 = \frac{T(n-T)}{n(n-1)}$ for Bernoulli data; both B and C are the same expression; $\bar{X}(1-\bar{X})$ uses n denominator (biased)

Q182. UMVUE Exponential Survival [HIGH]

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$, UMVUE of $S(t) = P(X > t) = e^{-\lambda t}$ is:

- A) $e^{-t/\bar{X}}$
- B) $(1-t/T)^{n-1} I(T > t)$ where $T = \sum X_i$
- C) $e^{-t\hat{\lambda}}$ where $\hat{\lambda} = 1/\bar{X}$
- D) $\frac{1}{n} \sum I(X_i > t)$

Answer: B

Trap/Note: $E[I(X_1 > t) | T = \sum X_i]$: given T , $X_1/T \sim \text{Beta}(1, n-1)$; $P(X_1 > t | T) = (1-t/T)^{n-1} I(T > t)$

Q183. MOM Weibull [HIGH]

For Weibull(k, λ) with both unknown, MOM requires:

- A) Solving two equations involving gamma functions: $E[X] = \lambda\Gamma(1+1/k)$, $E[X^2] = \lambda^2\Gamma(1+2/k)$
- B) $\hat{k} = \bar{X}/S$, $\hat{\lambda} = \bar{X}$
- C) $\hat{k} = 1/\log \bar{X}$, $\hat{\lambda} = e^{\bar{X}}$
- D) Closed form exists: $\hat{k} = \sqrt{6}S/(\pi\bar{X})$

Answer: A

Trap/Note: Weibull MOM: two equations with gamma functions; ratio $E[X^2]/(E[X])^2 = \Gamma(1+2/k)/[\Gamma(1+1/k)]^2$ depends only on k ; solve numerically

Q184. UMVUE Normal Sigma Unknown Mu [HIGH]

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with both unknown, UMVUE of σ is:

- A) $S = \sqrt{S^2}$
- B) S/c_4 where $c_4 = \sqrt{2/(n-1)} \cdot \Gamma(n/2)/\Gamma((n-1)/2)$

- C) $S \cdot \sqrt{(n-1)/2} \cdot \Gamma((n-1)/2)/\Gamma(n/2)$
- D) Both B and C (equivalent)

Answer: D

Trap/Note: $E[S] = \sigma c_4$; UMVUE of σ is S/c_4 ; $c_4 < 1$ so UMVUE $> S$; for large n , $c_4 \approx 1$

Q185. Rao Blackwell Geometric [HIGH]

For $X_1, \dots, X_n \sim \text{Geometric}(p)$, improve $T_0 = I(X_1 = 1)$ using $S = \sum X_i$:

- A) $E[I(X_1 = 1)|S = s] = \binom{s-1}{n-1}/\binom{s}{n}$ (negative hypergeometric)
- B) $E[I(X_1 = 1)|S = s] = p$
- C) $E[I(X_1 = 1)|S = s] = n/s$
- D) $E[I(X_1 = 1)|S = s] = (n-1)/(s-1)$

Answer: C

Trap/Note: Given $\sum X_i = s$, by symmetry $P(X_1 = 1|S = s) = P(X_j = 1|S = s)$ for all j ; sums to $n \cdot P(X_1 = 1|S = s)$; careful computation gives n/s

Q186. UMVUE Pareto [HIGH]

For $X_1, \dots, X_n \sim \text{Pareto}(\theta)$, $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, UMVUE of θ is:

- A) $-n/\sum \log X_i$
- B) $-(n-1)/\sum \log X_i$
- C) $n/\sum \log X_i$
- D) $(n+1)/\sum \log X_i$

Answer: B

Trap/Note: $T = \sum \log X_i$; $-\log X_i \sim \text{Exp}(\theta)$; $T \sim \Gamma(n, 1/\theta)$; $E[1/T] = \theta/(n-1)$; UMVUE of θ : $E[-(n-1)/T] = \theta$ ✓

Q187. MOM Inverse Gaussian [HIGH]

For $X \sim IG(\mu, \lambda)$, MOM estimators:

- A) $\hat{\mu} = \bar{X}$, $\hat{\lambda} = \bar{X}^3/S^2$
- B) $\hat{\mu} = \bar{X}$, $\hat{\lambda} = \bar{X}^2/S^2$
- C) $\hat{\mu} = \bar{X}$, $\hat{\lambda} = S^2/\bar{X}$
- D) $\hat{\mu} = 1/\bar{X}$, $\hat{\lambda} = \bar{X}^2/S^2$

Answer: A

Trap/Note: $E[X] = \mu$, $\text{Var}(X) = \mu^3/\lambda$; $\hat{\mu} = \bar{X}$; $\hat{\lambda} = \hat{\mu}^3/S^2 = \bar{X}^3/S^2$

Q188. UMVUE Hypergeometric [HIGH]

For Hypergeometric(N, M, n) with N known, UMVUE of M/N (population proportion) is:

- A) X/n (sample proportion)
- B) X/N
- C) $(X/n) \cdot N/(N-1)$

- D) X/n is already UMVUE

Answer: D

Trap/Note: X/n is unbiased for M/N and is a function of the complete sufficient statistic X ; hence it is UMVUE

Q189. MOM Truncated Normal [HIGH]

For $N(\mu, 1)$ truncated to $(0, \infty)$, MOM estimator of μ satisfies:

- A) $\bar{X} = \mu + \phi(\mu)/\Phi(\mu)$ (Mills ratio equation)
- B) $\hat{\mu} = \bar{X}$
- C) $\hat{\mu} = \bar{X} - 1$
- D) $\hat{\mu} = \bar{X} \cdot \Phi(\bar{X})$

Answer: A

Trap/Note: $E[X|X > 0] = \mu + \phi(\mu)/\Phi(\mu)$; setting equal to $\bar{X}$ gives implicit equation; same as MLE equation for truncated normal

Q190. UMVUE Double Exponential [HIGH]

For $X_1, \dots, X_n \sim \text{Laplace}(\theta, 1)$, UMVUE of θ :

- A) Is sample median
- B) Is $\bar{X}$
- C) Does not exist (no complete sufficient statistic)
- D) Is a weighted average of order statistics

Answer: C

Trap/Note: Laplace is a location family; order statistics are minimal sufficient but NOT complete; no UMVUE exists; sample median is best equivariant

Q191. Rao Blackwell Normal [HIGH]

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, improve $T_0 = X_1^2 - \sigma^2$ (unbiased for μ^2) using $S = \bar{X}$:

- A) $\bar{X}^2 - \sigma^2/n$
- B) $\bar{X}^2 - \sigma^2$
- C) $\bar{X}^2$
- D) $\bar{X}^2 + \sigma^2/n$

Answer: A

Trap/Note: $E[X_1^2 - \sigma^2 | \bar{X}] = E[X_1^2 | \bar{X}] - \sigma^2$; $E[X_1^2 | \bar{X}] = \bar{X}^2 + \sigma^2(n-1)/n$; so result = $\bar{X}^2 + \sigma^2(n-1)/n - \sigma^2 = \bar{X}^2 - \sigma^2/n$ ✓

Q192. UMVUE Binomial Probability [HIGH]

For $X_1, \dots, X_n \sim \text{Bin}(m, p)$ with m known, UMVUE of $P(X = k) = \binom{m}{k} p^k (1-p)^{m-k}$ is:

- A) $\binom{m}{k} \hat{p}^k (1-\hat{p})^{m-k}$ where $\hat{p} = \bar{X}/m$
- B) $\binom{nm-T}{m-k} \binom{T}{k} / \binom{nm}{m}$ where $T = \sum X_i$

- C) $\frac{\binom{m}{k}\binom{nm-m}{T-k}}{\binom{nm}{T}}$ where $T = \sum X_i$
- D) Does not exist in closed form

Answer: C

Trap/Note: Hypergeometric-type conditional probability; $E[I(X_1 = k)|T = \sum X_i]$ computed via hypergeometric distribution; C is the correct form

Q193. MOM Generalized Pareto [HIGH]

For Generalized Pareto $GP(\xi, \sigma)$, MOM estimators:

- A) $\hat{\xi} = 1 - \bar{X}^2/(2S^2)$, $\hat{\sigma} = \bar{X}(1 - \hat{\xi})$ (Hosking-Wallis)
- B) $\hat{\xi} = \bar{X}/S$, $\hat{\sigma} = S$
- C) $\hat{\xi} = S/\bar{X} - 1$, $\hat{\sigma} = \bar{X}$
- D) $\hat{\xi} = 1/2 - \bar{X}^2/S^2$, $\hat{\sigma} = \bar{X}(1 + \hat{\xi})$

Answer: A

Trap/Note: $E[X] = \sigma/(1 - \xi)$, $\text{Var}(X) = \sigma^2/[(1 - \xi)^2(1 - 2\xi)]$; solving gives Hosking-Wallis MOM; valid only for $\xi < 1/2$

Q194. UMVUE Geometric Probability [HIGH]

For $X_1, \dots, X_n \sim \text{Geometric}(p)$, UMVUE of p is:

- A) $1/\bar{X}$
- B) $(n-1)/(\sum X_i - 1)$
- C) $n/\sum X_i$
- D) $1/(\bar{X} - 1)$

Answer: B

Trap/Note: $T = \sum X_i \sim \text{NegBin}(n, p)$; $E[T] = n/p$; unbiased estimator: n/T is NOT exactly unbiased; correct UMVUE: $(n-1)/(T-1)$ since $E[(n-1)/(T-1)] = p$ for NegBin

Q195. Rao Blackwell Uniform [HIGH]

For $X_1, \dots, X_n \sim U(0, \theta)$, improve $T_0 = 2X_1$ (unbiased for θ) using $S = X_{(n)}$:

- A) $E[2X_1|X_{(n)} = t] = \frac{n+1}{n}t$
- B) $E[2X_1|X_{(n)} = t] = t$
- C) $E[2X_1|X_{(n)} = t] = \frac{2t}{n+1}$
- D) $E[2X_1|X_{(n)} = t] = \frac{(n+1)t}{n}$

Answer: A

Trap/Note: Given $X_{(n)} = t$: $E[X_1|X_{(n)} = t] = \frac{1}{n}t + \frac{n-1}{n} \cdot \frac{t}{2} = \frac{t(n+1)}{2n}$; so $E[2X_1|X_{(n)} = t] = \frac{(n+1)t}{n}$
✓

Q196. MOM Compound Poisson [HIGH]

For compound Poisson where $N \sim \text{Poisson}(\lambda)$ and $Y = \sum_{i=1}^N X_i$ with $X_i \sim \text{Exp}(\mu)$, MOM for (λ, μ) :

- A) $E[Y] = \lambda/\mu$, $\text{Var}(Y) = 2\lambda/\mu^2$ (use $E[Y] = \lambda E[X]$, $\text{Var}(Y) = \lambda E[X^2]$)
- B) $\hat{\lambda} = \bar{Y}\hat{\mu}$, $\hat{\mu} = \bar{Y}/S^2$ solve simultaneously
- C) $\hat{\mu} = 2\bar{Y}/S^2$, $\hat{\lambda} = \bar{Y}\hat{\mu}$
- D) $\hat{\mu} = \sqrt{2\bar{Y}/S^2}$, $\hat{\lambda} = \bar{Y}\hat{\mu}$

Answer: C

Trap/Note: $E[Y] = \lambda/\mu$, $\text{Var}(Y) = \lambda E[X^2] = \lambda \cdot 2/\mu^2$; ratio: $\text{Var}/E = 2/\mu$; $\hat{\mu} = 2\bar{Y}/S^2$; then $\hat{\lambda} = \bar{Y}\hat{\mu}$

Q197. UMVUE Normal Quantile [HIGH]

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, UMVUE of $\xi_p = \mu + z_p\sigma$ (the p -th quantile) is:

- A) $\bar{X} + z_p\sigma$
- B) $\bar{X} + z_p\sigma/\sqrt{n}$
- C) $\bar{X} + z_pS$
- D) $\bar{X} + z_p\sigma\sqrt{1 + 1/n}$

Answer: A

Trap/Note: $\bar{X}$ is UMVUE of μ ; $z_p\sigma$ is a constant; UMVUE of $\mu + z_p\sigma$ is $\bar{X} + z_p\sigma$; note σ is KNOWN here

Q198. MOM Mixture Normal [HIGH]

For mixture $\pi N(\mu_1, \sigma^2) + (1 - \pi)N(\mu_2, \sigma^2)$ with σ^2 known, MOM requires:

- A) At least 3 moment equations for 3 unknowns (π, μ_1, μ_2)
- B) 2 moment equations
- C) EM algorithm (MOM not applicable)
- D) Only first moment equation

Answer: A

Trap/Note: 3 unknown parameters need 3 equations; use $E[X]$, $E[X^2]$, $E[X^3]$; mixture MOM can have multiple solutions — EM preferred

Q199. UMVUE Negative Binomial [HIGH]

For $X_1, \dots, X_n \sim \text{NegBin}(r, p)$ with r known, UMVUE of p is:

- A) $nr / \sum X_i$
- B) $nr / (\sum X_i + nr)$
- C) $(nr - 1) / (\sum X_i + nr - 1)$
- D) $r / (\bar{X} + r)$

Answer: C

Trap/Note: $T = \sum X_i \sim \text{NegBin}(nr, p)$; UMVUE of p : $E[(nr - 1) / (T + nr - 1)] = p$ (analogous to geometric case); careful with exact formula

Q200. MOM Generalized Extreme Value [HIGH]

For $\text{GEV}(\mu, \sigma, \xi)$, MOM estimators:

- **A)** Exist in closed form for all ξ
- **B)** Exist only for $\xi < 1/3$ (finite third moment required)
- **C)** Require $\xi < 1/2$ (finite variance)
- **D)** Are always less efficient than MLE

Answer: B

Trap/Note: GEV moments: $E[X]$ exists for $\xi < 1$; $\text{Var}(X)$ for $\xi < 1/2$; third moment for $\xi < 1/3$; MOM using first 3 moments requires $\xi < 1/3$

Q201–Q250. Confidence Intervals

Q201. CI Normal Mean [MEDIUM]

100(1 - α)% CI for μ when σ^2 known is:

- A) $\bar{X} \pm z_{\alpha/2}\sigma/\sqrt{n}$
- B) $\bar{X} \pm z_{\alpha/2}\sigma$
- C) $\bar{X} \pm t_{\alpha/2, n-1}S/\sqrt{n}$
- D) $\bar{X} \pm z_{\alpha/2}S/\sqrt{n}$

Answer: A

Trap/Note: Known σ : use $z_{\alpha/2}$; unknown σ : use t_{n-1} ; never mix z with S or t with σ

Q202. CI Normal Variance [MEDIUM]

100(1 - α)% CI for σ^2 is:

- A) $\left[\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2} \right]$
- B) $\left[\frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2}, \frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2} \right]$
- C) $\left[S^2 \pm z_{\alpha/2}\sqrt{2\sigma^4/n} \right]$
- D) $\left[\frac{nS^2}{\chi_{\alpha/2, n}^2}, \frac{nS^2}{\chi_{1-\alpha/2, n}^2} \right]$

Answer: B

Trap/Note: χ^2 CI: smaller χ^2 value in denominator gives LARGER bound; lower bound uses LARGER χ^2 quantile; $(n - 1)$ df not n

Q203. CI Proportion [MEDIUM]

Wald CI for proportion p is:

- A) $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$
- B) $\hat{p} \pm z_{\alpha/2}\sqrt{p(1-p)/n}$
- C) $\hat{p} \pm t_{\alpha/2, n-1}\sqrt{\hat{p}(1-\hat{p})/n}$
- D) $\hat{p} \pm z_{\alpha/2}/\sqrt{n}$

Answer: A

Trap/Note: Wald uses estimated $\hat{p}$ in SE; Wilson CI is preferred for small n ; B uses true p (unknown); never use t for proportions

Q204. CI Shortest [HIGH]

For symmetric unimodal distributions, shortest CI is:

- A) Equal-tails CI
- B) Highest density region (HDR)
- C) One-sided CI
- D) Depends on sample size

Answer: B

Trap/Note: For symmetric distributions, equal-tails = HDR; for skewed (e.g., χ^2), HDR is shorter than equal-tails; equal-tails is NOT always shortest

Q205. CI Pivotal Quantity [MEDIUM]

A pivotal quantity $Q(X, \theta)$ has distribution:

- A) That depends on θ
- B) That does not depend on θ
- C) That is always normal
- D) That is always χ^2

Answer: B

Trap/Note: Pivot: distribution free of θ ; examples: $(\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)$; $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$

Q206. CI Exponential [MEDIUM]

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$, exact CI for λ uses pivot:

- A) $2\lambda \sum X_i \sim \chi_{2n}^2$
- B) $\lambda \sum X_i \sim \chi_n^2$
- C) $\sqrt{n}(\bar{X} - 1/\lambda) \sim N(0, 1/\lambda^2)$
- D) $n\lambda\bar{X} \sim \chi_n^2$

Answer: A

Trap/Note: $2\lambda X_i \sim \chi_2^2$; sum of n independent χ_2^2 is χ_{2n}^2 ; CI: $[\chi_{1-\alpha/2, 2n}^2/(2 \sum X_i), \chi_{\alpha/2, 2n}^2/(2 \sum X_i)]$

Q207. CI Uniform [HIGH]

For $X_1, \dots, X_n \sim U(0, \theta)$, exact one-sided CI for θ based on $X_{(n)}$ is:

- A) $[X_{(n)}, X_{(n)} \cdot \alpha^{-1/n}]$
- B) $[X_{(n)}, X_{(n)} \cdot \alpha^{1/n}]$
- C) $[X_{(n)} \cdot \alpha^{1/n}, \infty)$
- D) $[X_{(n)}/\alpha^{1/n}, \infty)$

Answer: A

Trap/Note: $U = X_{(n)}/\theta \sim \text{Beta}(n, 1)$; $P(U \geq \alpha^{1/n}) = 1 - \alpha$; so $P(\theta \leq X_{(n)}/\alpha^{1/n}) = 1 - \alpha$; one-sided CI: $[X_{(n)}, X_{(n)}\alpha^{-1/n}]$

Q208. CI Fiducial [HIGH]

Fiducial interval (Fisher) differs from confidence interval in that:

- A) It treats θ as random given observed data
- B) It has different coverage probability
- C) It requires prior distribution
- D) It is always shorter

Answer: A

Trap/Note: Fiducial: treats parameter as having distribution induced by data (without prior); numerically equals CI for location families; philosophically different from frequentist CI

Q209. CI Bayesian Credible [HIGH]

A 95% Bayesian credible interval differs from 95% frequentist CI in that:

- A) $P(\theta \in \text{CI}|X) = 0.95$ (posterior probability)
- B) $P(\theta \in \text{CI}) = 0.95$ (prior probability)
- C) Coverage is 95% in repeated sampling
- D) It is always wider

Answer: A

Trap/Note: Bayesian: posterior probability statement about θ ; frequentist: coverage in repeated sampling; C is the frequentist interpretation; A is Bayesian

Q210. CI Wilson Proportion [HIGH]

Wilson CI for proportion p is centered at:

- A) $\hat{p}$
- B) $\frac{\hat{p} + z_{\alpha/2}^2/(2n)}{1 + z_{\alpha/2}^2/n}$ (adjusted center)
- C) p (true proportion)
- D) $1/2$

Answer: B

Trap/Note: Wilson CI: add $z^2/(2n)$ pseudo-observations; center shifts toward $1/2$; better coverage than Wald for small n or extreme p

Q211. CI Prediction Interval [MEDIUM]

Prediction interval for future observation $X_{n+1} \sim N(\mu, \sigma^2)$ with σ^2 known is:

- A) $\bar{X} \pm z_{\alpha/2}\sigma\sqrt{1 + 1/n}$
- B) $\bar{X} \pm z_{\alpha/2}\sigma/\sqrt{n}$
- C) $\bar{X} \pm z_{\alpha/2}\sigma$
- D) $\bar{X} \pm t_{\alpha/2, n-1}S\sqrt{1 + 1/n}$

Answer: A

Trap/Note: Prediction interval wider than CI: $X_{n+1} - \bar{X} \sim N(0, \sigma^2(1 + 1/n))$; D is correct when σ unknown; A when σ known

Q212. CI Tolerance Interval [HIGH]

A tolerance interval $[L, U]$ with content p and confidence $1 - \alpha$ means:

- A) $P(P(L \leq X \leq U) \geq p) \geq 1 - \alpha$
- B) $P(L \leq X \leq U) = p$
- C) $P(L \leq \mu \leq U) = 1 - \alpha$
- D) $P(L \leq X_{n+1} \leq U) = p$

Answer: A

Trap/Note: Tolerance interval: probabilistic statement about coverage of future population; B is exact (not achievable for continuous distributions); C is CI for mean; D is prediction interval

Q213. CI Bootstrap Percentile [MEDIUM]

Bootstrap percentile CI uses:

- A) $[\hat{\theta}_{(\alpha/2)}^*, \hat{\theta}_{(1-\alpha/2)}^*]$ — quantiles of bootstrap distribution
- B) $[2\hat{\theta} - \hat{\theta}_{(1-\alpha/2)}^*, 2\hat{\theta} - \hat{\theta}_{(\alpha/2)}^*]$ — reflected
- C) $\hat{\theta} \pm z_{\alpha/2} \text{SE}_{\text{boot}}$
- D) $\hat{\theta} \pm t_{\alpha/2, n-1} \text{SE}_{\text{boot}}$

Answer: A

Trap/Note: Percentile method: A; Basic/Reverse percentile: B; Normal bootstrap: C; BCa (bias-corrected accelerated) is most accurate

Q214. CI Bootstrap BCa [HIGH]

BCa bootstrap CI corrects for:

- A) Bias and skewness of bootstrap distribution
- B) Only bias
- C) Only skewness
- D) Non-normality of original data

Answer: A

Trap/Note: BCa = Bias-Corrected and accelerated; bias correction $\hat{z}_0$ and acceleration $\hat{a}$ both applied; most accurate bootstrap CI method

Q215. CI Jackknife SE [MEDIUM]

Jackknife standard error estimate is:

- A) $\sqrt{\frac{n-1}{n} \sum (\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)})^2}$
- B) $\sqrt{\frac{1}{n} \sum (\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)})^2}$
- C) $\sqrt{\frac{n-1}{n^2} \sum (\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)})^2}$
- D) $\sqrt{\frac{n-1}{n} \sum (\hat{\theta}_{(i)} - \hat{\theta})^2}$

Answer: A

Trap/Note: Jackknife SE = $\sqrt{\frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)})^2}$; the $(n-1)/n$ factor (not $1/n$) is crucial

Q216. CI Optimum Unbiased [HIGH]

Optimum confidence interval (shortest unbiased) for μ in $N(\mu, \sigma^2)$ with σ^2 unknown is:

- A) Equal-tails t -interval

- B) One-sided t -interval
- C) z -interval using S
- D) Shortest t -interval (not equal-tails)

Answer: A

Trap/Note: For t -distribution (symmetric), equal-tails interval IS the shortest unbiased interval; for asymmetric distributions (e.g., χ^2), equal-tails is NOT shortest

Q217. CI Ratio Two Means [HIGH]

CI for ratio μ_1/μ_2 of two normal means uses:

- A) Fieller's theorem
- B) Delta method approximation
- C) Log transformation then back-transform
- D) All of the above are valid approaches

Answer: D

Trap/Note: Fieller's theorem gives exact CI; delta method gives approximate CI; log transformation is another approximation; all are used in practice

Q218. CI Nonparametric Median [HIGH]

Distribution-free CI for population median uses:

- A) Order statistics $X_{(r)}$ and $X_{(s)}$ with coverage from binomial distribution
- B) $\bar{X} \pm z_{\alpha/2}S/\sqrt{n}$
- C) Sample median $\pm z_{\alpha/2} \cdot \pi/(2n)$ (asymptotic)
- D) Bootstrap only

Answer: A

Trap/Note: Exact nonparametric CI for median: $[X_{(r)}, X_{(s)}]$ where r, s chosen so $P(r \leq \text{Bin}(n, 1/2) \leq s-1) \geq 1 - \alpha$; C is asymptotic approximation

Q219. CI Simultaneous Bonferroni [MEDIUM]

Bonferroni simultaneous CIs for k parameters at family-wise level $1 - \alpha$ use individual level:

- A) $1 - \alpha/k$
- B) $1 - \alpha/2$
- C) $1 - k\alpha$
- D) $(1 - \alpha)^{1/k}$

Answer: A

Trap/Note: Bonferroni: use α/k per interval; D is Šidák correction (exact for independent tests); both are conservative

Q220. CI Bootstrap Double [HIGH]

Double bootstrap (bootstrap-t) CI:

- **A)** Uses bootstrap to estimate the distribution of $(\hat{\theta} - \theta)/\hat{SE}$
- **B)** Applies bootstrap twice to reduce bias
- **C)** Uses t -distribution with bootstrap SE
- **D)** Is equivalent to BCa

Answer: A

Trap/Note: Bootstrap-t: estimate distribution of studentized statistic via nested bootstrap; computationally expensive but second-order accurate; different from BCa

Q221. CI Profile Likelihood [HIGH]

Profile likelihood CI for ψ is:

- **A)** $\{\psi : -2[\ell_p(\psi) - \ell_p(\hat{\psi})] \leq \chi_{\alpha,1}^2\}$
- **B)** $\hat{\psi} \pm z_{\alpha/2}/\sqrt{nI(\hat{\psi})}$
- **C)** $\{\psi : \ell_p(\psi) \geq \ell_p(\hat{\psi}) - z_{\alpha/2}^2/2\}$
- **D)** Both A and C (equivalent since $\chi_{\alpha,1}^2 = z_{\alpha/2}^2$)

Answer: D

Trap/Note: $\chi_{0.05,1}^2 = 3.841 = 1.96^2$; A and C are equivalent; profile likelihood CI is second-order accurate unlike Wald CI

Q222. CI Exponential One Sided [MEDIUM]

One-sided 95% upper CI for λ in $\text{Exp}(\lambda)$ based on n observations:

- **A)** $\lambda \leq \chi_{0.05,2n}^2/(2 \sum X_i)$
- **B)** $\lambda \leq \chi_{0.95,2n}^2/(2 \sum X_i)$
- **C)** $\lambda \leq z_{0.05}/(\bar{X}\sqrt{n})$
- **D)** $\lambda \leq \chi_{0.05,2n}^2/\sum X_i$

Answer: A

Trap/Note: $P(2\lambda \sum X_i \leq \chi_{0.05,2n}^2) = 0.05$; upper CI: $\lambda \leq \chi_{0.05,2n}^2/(2 \sum X_i)$; note $\chi_{0.05}^2$ is LOWER tail quantile giving UPPER bound for λ

Q223. CI Poisson Exact [HIGH]

Exact CI for Poisson rate λ based on observed count x is:

- **A)** $[\chi_{\alpha/2,2x}^2/(2n), \chi_{1-\alpha/2,2(x+1)}^2/(2n)]$
- **B)** $[\bar{X} \pm z_{\alpha/2}\sqrt{\bar{X}/n}]$
- **C)** $[x/n \pm z_{\alpha/2}\sqrt{x/n}]$
- **D)** $[\chi_{1-\alpha/2,2x}^2/(2n), \chi_{\alpha/2,2(x+1)}^2/(2n)]$

Answer: A

Trap/Note: Exact Poisson CI: lower = $\chi^2_{\alpha/2, 2x}/2$ (for rate per unit), upper = $\chi^2_{1-\alpha/2, 2(x+1)}/2$; divide by n for rate; careful with which tail goes where

Q224. CI Correlation [HIGH]

Fisher's z -transformation CI for correlation ρ :

- A) $\tanh(\tanh^{-1}(\hat{\rho}) \pm z_{\alpha/2}/\sqrt{n-3})$
- B) $\hat{\rho} \pm z_{\alpha/2} \sqrt{(1-\hat{\rho}^2)^2/(n-1)}$
- C) $\hat{\rho} \pm z_{\alpha/2}/\sqrt{n}$
- D) $\hat{\rho} \pm t_{\alpha/2, n-2} \sqrt{(1-\hat{\rho}^2)/(n-2)}$

Answer: A

Trap/Note: Fisher z : $z = \tanh^{-1}(\hat{\rho}) = \frac{1}{2} \log \frac{1+\hat{\rho}}{1-\hat{\rho}}$; $z \sim N(\tanh^{-1}(\rho), 1/(n-3))$; back-transform via $\tanh$

Q225. CI Difference Proportions [MEDIUM]

CI for $p_1 - p_2$ (two independent proportions):

- A) $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}$
- B) $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}$ where $\hat{p}$ is pooled
- C) A for CI; B for hypothesis test
- D) Both A and B are valid CIs

Answer: C

Trap/Note: For CI: use unpooled SE (A); for hypothesis test $H_0 : p_1 = p_2$: use pooled SE (B); common exam trap to mix them

Q226. CI Regression Slope [MEDIUM]

CI for slope β in simple linear regression:

- A) $\hat{\beta} \pm t_{\alpha/2, n-2} \cdot S/\sqrt{S_{xx}}$ where $S_{xx} = \sum(x_i - \bar{x})^2$
- B) $\hat{\beta} \pm t_{\alpha/2, n-1} \cdot S/\sqrt{S_{xx}}$
- C) $\hat{\beta} \pm z_{\alpha/2} \cdot \sigma/\sqrt{S_{xx}}$
- D) $\hat{\beta} \pm t_{\alpha/2, n-2} \cdot S_{\hat{\beta}}$ where $S_{\hat{\beta}}^2 = S^2/S_{xx}$

Answer: D

Trap/Note: A and D are equivalent; $n-2$ df (not $n-1$); $S^2 = \text{RSS}/(n-2)$; C uses known σ

Q227. CI Bootstrap Parametric [HIGH]

Parametric bootstrap CI:

- A) Resamples from fitted parametric model $f(x; \hat{\theta})$
- B) Resamples from empirical distribution
- C) Uses analytical formula with bootstrap SE
- D) Is equivalent to nonparametric bootstrap

Answer: A

Trap/Note: Parametric bootstrap: simulate from $f(x; \hat{\theta})$; nonparametric: resample from data; they differ when model is misspecified

Q228. CI Jackknife Bias Corrected [HIGH]

Jackknife bias-corrected estimator is:

- **A)** $n\hat{\theta} - (n-1)\hat{\theta}_{(\cdot)}$
- **B)** $\hat{\theta} - (n-1)(\hat{\theta}_{(\cdot)} - \hat{\theta})$
- **C)** $\hat{\theta} + \text{bias}_{\text{jack}}$ where $\text{bias} = (n-1)(\hat{\theta}_{(\cdot)} - \hat{\theta})$
- **D)** All of the above (equivalent)

Answer: D

Trap/Note: Jackknife bias estimate = $(n-1)(\hat{\theta}_{(\cdot)} - \hat{\theta})$; bias-corrected = $\hat{\theta} - \text{bias} = \hat{\theta} - (n-1)(\hat{\theta}_{(\cdot)} - \hat{\theta}) = n\hat{\theta} - (n-1)\hat{\theta}_{(\cdot)}$; all equivalent

Q229. CI Subsampling [HIGH]

Subsampling CI differs from bootstrap in that:

- **A)** It samples WITHOUT replacement (subsets of size $b < n$)
- **B)** It requires correct specification of convergence rate
- **C)** It is valid under weaker conditions than bootstrap
- **D)** All of the above

Answer: D

Trap/Note: Subsampling: without replacement, size $b \rightarrow \infty$ with $b/n \rightarrow 0$; valid even when bootstrap fails (e.g., for maximum of i.i.d. uniform); requires knowing rate

Q230. CI Simultaneous Scheffe [HIGH]

Scheffé simultaneous CI for all linear contrasts $c^T \beta$ in regression:

- **A)** $\hat{c}^T \hat{\beta} \pm \sqrt{p \cdot F_{\alpha, p, n-p}} \cdot \text{SE}$
- **B)** $\hat{c}^T \hat{\beta} \pm t_{\alpha/2, n-p} \cdot \text{SE}$
- **C)** $\hat{c}^T \hat{\beta} \pm z_{\alpha/2} \cdot \text{SE}$
- **D)** $\hat{c}^T \hat{\beta} \pm \sqrt{F_{\alpha, 1, n-p}} \cdot \text{SE}$

Answer: A

Trap/Note: Scheffé: $\sqrt{p \cdot F}$ critical value covers ALL contrasts simultaneously; Bonferroni covers only pre-specified contrasts; p = number of parameters

Q231. CI Nonparametric Bootstrap Validity [HIGH]

Bootstrap CI is NOT valid when:

- **A)** Distribution has heavy tails
- **B)** Statistic is the sample maximum from $U(0, \theta)$

- C) Sample size is small
- D) Distribution is skewed

Answer: B

Trap/Note: Bootstrap fails for non-regular statistics like sample maximum (convergence rate n^{-1} not $n^{-1/2}$); subsampling or m -out-of- n bootstrap needed

Q232. CI Likelihood Ratio [HIGH]

Likelihood ratio CI has the property of being:

- A) Transformation invariant (equivariant under reparametrization)
- B) Always equal to Wald CI
- C) Always shorter than Wald CI
- D) Exact for all distributions

Answer: A

Trap/Note: LR CI is transformation invariant (key advantage over Wald); Wald CI is NOT invariant; LR CI is generally more accurate than Wald in finite samples

Q233. CI Bootstrap Iterations [MEDIUM]

For bootstrap CI, recommended number of bootstrap replications B :

- A) $B = 100$ is always sufficient
- B) $B \geq 1000$ for percentile CI; $B \geq 2000$ for BCa
- C) $B = n$ (same as sample size)
- D) $B = \infty$ (theoretical limit)

Answer: B

Trap/Note: Practical guidelines: $B = 1000$ for SE and percentile; $B = 2000$ -5000 for BCa; more replications reduce Monte Carlo error in CI endpoints

Q234. CI Exact Binomial [HIGH]

Clopper-Pearson exact CI for p based on x successes in n trials:

- A) $[B_{\alpha/2;x,n-x+1}, B_{1-\alpha/2;x+1,n-x}]$ where B is Beta quantile
- B) $[\hat{p} \pm z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}]$
- C) $[F_{\alpha/2,2x,2(n-x+1)} \cdot x/(n-x+1 + F \cdot x), \dots]$
- D) Both A and C (equivalent via Beta-F relationship)

Answer: D

Trap/Note: Clopper-Pearson uses Beta distribution; equivalent F-distribution form also exists; both are exact (conservative); B is Wald (approximate)

Q235. CI Regression Prediction [MEDIUM]

Prediction interval for new response Y_0 at x_0 in regression:

- A) $\hat{Y}_0 \pm t_{\alpha/2, n-2} S \sqrt{1 + 1/n + (x_0 - \bar{x})^2 / S_{xx}}$

- B) $\hat{Y}_0 \pm t_{\alpha/2, n-2} S \sqrt{1/n + (x_0 - \bar{x})^2 / S_{xx}}$
- C) $\hat{Y}_0 \pm t_{\alpha/2, n-2} S / \sqrt{n}$
- D) $\hat{Y}_0 \pm z_{\alpha/2} \sigma \sqrt{1 + 1/n + (x_0 - \bar{x})^2 / S_{xx}}$

Answer: A

Trap/Note: Prediction interval: extra 1 under square root (for new observation variance); B is CI for mean response; A is wider than B

Q236. CI Variance Ratio [MEDIUM]

CI for σ_1^2 / σ_2^2 (two independent normal samples):

- A) $[S_1^2 / (S_2^2 \cdot F_{\alpha/2, n_1-1, n_2-1}), S_1^2 \cdot F_{\alpha/2, n_2-1, n_1-1} / S_2^2]$
- B) $[S_1^2 / S_2^2 \pm z_{\alpha/2} \sqrt{2/(n_1 - 1) + 2/(n_2 - 1)}]$
- C) $[(S_1^2 / S_2^2) / F_{1-\alpha/2, n_1-1, n_2-1}, (S_1^2 / S_2^2) \cdot F_{1-\alpha/2, n_2-1, n_1-1}]$
- D) Both A and C (equivalent)

Answer: D

Trap/Note: $F_{1-\alpha/2, m, n} = 1 / F_{\alpha/2, n, m}$; both forms are equivalent; note df order reversal in the reciprocal

Q237. CI Bootstrap Studentized [HIGH]

Studentized bootstrap (bootstrap-t) CI is:

- A) Second-order accurate (error $O(n^{-1})$)
- B) First-order accurate (error $O(n^{-1/2})$)
- C) Exact
- D) Less accurate than percentile bootstrap

Answer: A

Trap/Note: Bootstrap-t is second-order accurate; percentile and basic bootstrap are first-order; BCa is also second-order; studentized bootstrap requires nested bootstrap for SE

Q238. CI Jackknife Failure [HIGH]

Jackknife fails (is inconsistent) for:

- A) Sample mean
- B) Sample median (non-smooth functional)
- C) Sample variance
- D) Regression coefficients

Answer: B

Trap/Note: Jackknife fails for non-smooth statistics like sample median, sample quantiles; bootstrap is preferred; jackknife works for smooth functionals

Q239. CI Simultaneous Tukey [HIGH]

Tukey's HSD simultaneous CI for pairwise differences $\mu_i - \mu_j$ in one-way ANOVA uses:

- A) Studentized range distribution $q_{\alpha,k,N-k}$
- B) t -distribution with Bonferroni correction
- C) F -distribution
- D) Normal distribution

Answer: A

Trap/Note: Tukey uses studentized range (maximum of pairwise t -statistics); exact for balanced designs; Bonferroni is conservative; Tukey is less conservative

Q240. CI Bootstrap Bias Correction [HIGH]

Bootstrap bias estimate for $\hat{\theta}$ is:

- A) $\bar{\theta}^* - \hat{\theta}$ where $\bar{\theta}^*$ is mean of bootstrap estimates
- B) $\hat{\theta} - \bar{\theta}^*$
- C) $E[\hat{\theta}] - \theta$
- D) $\bar{\theta}^* - \theta$

Answer: A

Trap/Note: Bootstrap bias = $E^*[\hat{\theta}^*] - \hat{\theta} = \bar{\theta}^* - \hat{\theta}$; bias-corrected estimate = $2\hat{\theta} - \bar{\theta}^*$; B has wrong sign

Q241. CI Fiducial Location [HIGH]

For location family $f(x - \theta)$, fiducial distribution of θ given $\bar{X} = \bar{x}$ is:

- A) $f(\bar{x} - \theta) \cdot \sqrt{n}$ (scaled)
- B) Posterior with flat prior
- C) Both A and B (coincide for location families)
- D) Normal distribution always

Answer: C

Trap/Note: For location families, fiducial = posterior with flat (improper) prior = confidence distribution; all three coincide — key result

Q242. CI Tolerance Normal [HIGH]

For $N(\mu, \sigma^2)$ with both unknown, tolerance interval covering at least proportion p with confidence $1 - \alpha$:

- A) $\bar{X} \pm k(n, p, \alpha)S$ where k is tolerance factor from tables
- B) $\bar{X} \pm z_p S$
- C) $\bar{X} \pm z_p \sigma$
- D) $\bar{X} \pm z_{p/2} S \sqrt{1 + 1/n}$

Answer: A

Trap/Note: Tolerance factor k accounts for estimation of both μ and σ ; $k > z_p$ always; $k \rightarrow z_p$ as $n \rightarrow \infty$; B undercovers

Q243. CI Bootstrap Smoothed [HIGH]

Smoothed bootstrap adds noise to resampled values to:

- A) Improve CI coverage for discrete distributions
- B) Reduce computation time
- C) Handle missing data
- D) Increase bias

Answer: A

Trap/Note: Smoothed bootstrap: add small random noise (bandwidth h) to discrete resamples; treats discrete data as continuous; improves coverage for discrete statistics

Q244. CI Confidence Distribution [HIGH]

A confidence distribution $C(\theta)$ satisfies:

- A) $C(\theta_0) \sim U(0, 1)$ under $H_0 : \theta = \theta_0$
- B) $P(\theta \in [C^{-1}(\alpha/2), C^{-1}(1 - \alpha/2)]) = 1 - \alpha$
- C) It is a proper probability distribution over θ
- D) All of the above

Answer: D

Trap/Note: Confidence distribution unifies CI, p -values, and point estimates; $C(\theta_0)$ is the p -value for $H_0 : \theta \leq \theta_0$; modern framework

Q245. CI Bootstrap Accelerated [HIGH]

The acceleration constant $\hat{a}$ in BCa bootstrap measures:

- A) Rate of change of SE with respect to θ
- B) Skewness of bootstrap distribution
- C) Bias of $\hat{\theta}$
- D) Sample size effect

Answer: A

Trap/Note: $\hat{a} = \frac{\sum(\hat{\theta}_{(i)} - \hat{\theta}_{(i)})^3}{6[\sum(\hat{\theta}_{(i)} - \hat{\theta}_{(i)})^2]^{3/2}}$ (jackknife-based); measures how SE changes with θ (influence function skewness)

Q246. CI Regression Simultaneous [HIGH]

Working-Hotelling simultaneous CI band for regression line $E[Y|x] = \alpha + \beta x$:

- A) $\hat{Y}(x) \pm \sqrt{2F_{\alpha, 2, n-2}} \cdot S\sqrt{1/n + (x - \bar{x})^2/S_{xx}}$
- B) $\hat{Y}(x) \pm t_{\alpha/2, n-2} \cdot S\sqrt{1/n + (x - \bar{x})^2/S_{xx}}$
- C) $\hat{Y}(x) \pm z_{\alpha/2} \cdot S\sqrt{1/n + (x - \bar{x})^2/S_{xx}}$
- D) $\hat{Y}(x) \pm F_{\alpha, 2, n-2} \cdot S\sqrt{1/n + (x - \bar{x})^2/S_{xx}}$

Answer: A

Trap/Note: Working-Hotelling: $\sqrt{2F}$ critical value for ALL x simultaneously; B is pointwise CI; $\sqrt{2F_{0.05,2,n-2}} > t_{0.025,n-2}$ so band is wider

Q247. CI Bootstrap Monte Carlo Error [HIGH]

Monte Carlo error in bootstrap CI endpoint $\hat{\theta}_{(\alpha)}^*$ is approximately:

- A) $\sqrt{\alpha(1-\alpha)/B} \cdot \hat{\sigma}^*$ where $\hat{\sigma}^*$ is bootstrap SD
- B) $1/\sqrt{B}$
- C) $\alpha/\sqrt{B}$
- D) $\hat{\sigma}^*/\sqrt{B}$

Answer: A

Trap/Note: SE of sample quantile: $\sqrt{p(1-p)/B}/f(\xi_p)$; for bootstrap distribution, $f(\xi_p) \approx 1/\hat{\sigma}^*$; so $SE \approx \hat{\sigma}^* \sqrt{\alpha(1-\alpha)/B}$

Q248. CI Empirical Likelihood [HIGH]

Empirical likelihood CI (Owen) is:

- A) Nonparametric, transformation invariant, Bartlett correctable
- B) Requires parametric model
- C) Always wider than bootstrap CI
- D) Equivalent to bootstrap percentile CI

Answer: A

Trap/Note: Empirical likelihood: nonparametric analog of likelihood ratio CI; shares properties of parametric LR CI (transformation invariance, Bartlett correction); Wilks theorem holds

Q249. CI Shortest Chi Square [HIGH]

Shortest CI for σ^2 (not equal-tails) requires finding χ_L^2, χ_U^2 satisfying:

- A) $\chi_{1-\alpha/2,n-1}^2 - \chi_{\alpha/2,n-1}^2$ minimized subject to coverage $1 - \alpha$
- B) $1/\chi_L^2 - 1/\chi_U^2$ minimized subject to $P(\chi_L^2 \leq \chi_{n-1}^2 \leq \chi_U^2) = 1 - \alpha$
- C) Equal tail probabilities
- D) χ_U^2/χ_L^2 minimized

Answer: B

Trap/Note: CI for σ^2 : $\text{length} \propto 1/\chi_L^2 - 1/\chi_U^2$; minimize this subject to coverage; NOT equal-tails; equal-tails is standard but not shortest

Q250. CI Bootstrap Double Coverage [HIGH]

Double bootstrap corrects coverage error of bootstrap CI from:

- A) $O(n^{-1/2})$ to $O(n^{-1})$
- B) $O(n^{-1})$ to $O(n^{-3/2})$
- C) $O(1)$ to $O(n^{-1/2})$
- D) Exact to approximate

Answer: A

Trap/Note: Single bootstrap CI: coverage error $O(n^{-1/2})$; double bootstrap: $O(n^{-1})$; studentized bootstrap also achieves $O(n^{-1})$; computationally expensive

Q251–Q310. Neyman–Pearson Lemma, UMP Tests & Power Functions

Q251. NP Lemma Statement [MEDIUM]

Neyman–Pearson lemma: Most powerful test of size α for $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ rejects when:

- A) $\Lambda = f(x; \theta_1)/f(x; \theta_0) > k$ where k chosen so $P_{H_0}(\Lambda > k) = \alpha$
- B) $\Lambda = f(x; \theta_0)/f(x; \theta_1) > k$
- C) $\bar{X} > k$ always
- D) $p\text{-value} < \alpha$

Answer: A

Trap/Note: Ratio is f_1/f_0 (alternative over null); reject for LARGE ratio; B has ratio inverted; k determined by size constraint

Q252. UMP Exponential Family [HIGH]

For one-parameter exponential family with monotone likelihood ratio in $T(x)$, UMP test of $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$:

- A) Rejects for large $T(x)$
- B) Rejects for small $T(x)$
- C) Rejects for $|T(x)| > k$
- D) Depends on direction of monotonicity

Answer: D

Trap/Note: If LR is increasing in T : reject for large T ; if decreasing: reject for small T ; must check direction

Q253. UMP Existence Two Sided [HIGH]

UMP test for two-sided $H_1 : \theta \neq \theta_0$ against one-sided $H_0 : \theta = \theta_0$:

- A) Always exists for exponential families
- B) Generally does NOT exist
- C) Exists only for normal distribution
- D) Exists when distribution is symmetric

Answer: B

Trap/Note: UMP for two-sided alternatives generally fails; the test that is MP against $\theta_1 > \theta_0$ differs from MP against $\theta_1 < \theta_0$; UMPU tests exist instead

Q254. NP Lemma Normal Mean [MEDIUM]

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, MP test of $H_0 : \mu = \mu_0$ vs $H_1 : \mu = \mu_1 > \mu_0$:

- A) Reject if $\bar{X} > c$ where $c = \mu_0 + z_\alpha \sigma / \sqrt{n}$
- B) Reject if $\bar{X} < c$
- C) Reject if $|\bar{X} - \mu_0| > c$
- D) Reject if $\bar{X} > \mu_1$

Answer: A

Trap/Note: $LR = \exp[n(\mu_1 - \mu_0)\bar{X}/\sigma^2 - \text{const}]$; increasing in $\bar{X}$ (since $\mu_1 > \mu_0$); reject for large $\bar{X}$; c from size constraint

Q255. Power Function Definition [MEDIUM]

Power function $\beta(\theta)$ of a test is:

- A) $P_\theta(\text{reject } H_0)$ for all θ
- B) $P_\theta(\text{accept } H_0)$ for $\theta \in H_1$
- C) $1 - P(\text{Type II error})$ only under H_1
- D) $P(\text{Type I error})$ under H_0

Answer: A

Trap/Note: Power function defined for ALL θ ; under H_0 : $\beta(\theta) = \alpha$ (size); under H_1 : $\beta(\theta) =$ power; C is incomplete

Q256. Critical Region Types [MEDIUM]

For testing $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$ (normal, σ known), critical region is:

- A) $\{|\bar{X} - \mu_0| > z_{\alpha/2}\sigma/\sqrt{n}\}$
- B) $\{\bar{X} > \mu_0 + z_\alpha\sigma/\sqrt{n}\}$
- C) $\{\bar{X} < \mu_0 - z_\alpha\sigma/\sqrt{n}\}$
- D) $\{(\bar{X} - \mu_0)^2 > z_\alpha^2\sigma^2/n\}$

Answer: A

Trap/Note: Two-sided test: A and D are equivalent; B is one-sided right; C is one-sided left; for two-sided, use $z_{\alpha/2}$ not z_α

Q257. Similar Region [HIGH]

A critical region C is similar of size α if:

- A) $P_\theta(X \in C) = \alpha$ for all $\theta \in H_0$
- B) $P_\theta(X \in C) \leq \alpha$ for all $\theta \in H_0$
- C) $P_\theta(X \in C) = \alpha$ for all θ
- D) $P_\theta(X \in C) \geq \alpha$ for all $\theta \in H_1$

Answer: A

Trap/Note: Similar region: EXACTLY α for all $\theta \in H_0$ (not just $\leq \alpha$); level α test has $\leq \alpha$; similar tests are useful for composite H_0

Q258. Unbiased Test [HIGH]

A test is unbiased if:

- A) $\beta(\theta) \geq \alpha$ for all $\theta \in H_1$ and $\beta(\theta) \leq \alpha$ for all $\theta \in H_0$
- B) $\beta(\theta) = \alpha$ for all θ
- C) Power equals size
- D) Type I error equals Type II error

Answer: A

Trap/Note: Unbiased: power under H_1 is at least as large as size; ensures test is not “worse than random”; B is similar test; D is unrelated

Q259. Randomized Test [HIGH]

A randomized test with critical function $\phi(x)$ means:

- **A)** Reject with probability $\phi(x) \in [0, 1]$
- **B)** Randomly choose between two tests
- **C)** Use random sample size
- **D)** Randomly permute data

Answer: A

Trap/Note: Randomized test: $\phi(x) = P(\text{reject}|X = x)$; needed to achieve exact size α for discrete distributions; $\phi(x) \in \{0, 1\}$ for non-randomized

Q260. NP Lemma Poisson [MEDIUM]

For $X \sim \text{Poisson}(\lambda)$, MP test of $H_0 : \lambda = \lambda_0$ vs $H_1 : \lambda = \lambda_1 > \lambda_0$:

- **A)** Reject for large X (i.e., $X \geq c$)
- **B)** Reject for small X
- **C)** Reject for $|X - \lambda_0| > c$
- **D)** Reject for $X = 0$

Answer: A

Trap/Note: LR = $(\lambda_1/\lambda_0)^x e^{-(\lambda_1-\lambda_0)}$; increasing in x (since $\lambda_1 > \lambda_0$); reject for large X ; randomization at boundary for exact size

Q261. UMP Normal Variance [HIGH]

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with μ known, UMP test of $H_0 : \sigma^2 \leq \sigma_0^2$ vs $H_1 : \sigma^2 > \sigma_0^2$:

- **A)** Reject for large $\sum (X_i - \mu)^2$
- **B)** Reject for small $\sum (X_i - \mu)^2$
- **C)** Reject for large $|\bar{X} - \mu|$
- **D)** UMP does not exist

Answer: A

Trap/Note: $\sum (X_i - \mu)^2 / \sigma^2 \sim \chi_n^2$; MLR in $\sum (X_i - \mu)^2$; reject for large values; μ known is crucial

Q262. Power Calculation Normal [MEDIUM]

For $H_0 : \mu = 0$ vs $H_1 : \mu = \delta > 0$ with σ known, power of z -test at level α is:

- **A)** $\Phi(-z_\alpha + \delta\sqrt{n}/\sigma)$
- **B)** $\Phi(z_\alpha - \delta\sqrt{n}/\sigma)$
- **C)** $1 - \Phi(z_\alpha - \delta\sqrt{n}/\sigma)$
- **D)** Both A and C (equivalent)

Answer: D

Trap/Note: Power = $P(\bar{X} > z_\alpha \sigma / \sqrt{n} | \mu = \delta) = P(Z > z_\alpha - \delta \sqrt{n} / \sigma) = 1 - \Phi(z_\alpha - \delta \sqrt{n} / \sigma) = \Phi(\delta \sqrt{n} / \sigma - z_\alpha)$; A and C equivalent

Q263. NP Lemma Exponential [MEDIUM]

For $X_1, \dots, X_n \sim \text{Exp}(\theta)$, MP test of $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1 < \theta_0$ (smaller rate = larger mean):

- **A)** Reject for large $\sum X_i$
- **B)** Reject for small $\sum X_i$
- **C)** Reject for large $\bar{X}$
- **D)** Both A and C (equivalent)

Answer: D

Trap/Note: LR = $(\theta_1 / \theta_0)^n \exp[-(\theta_1 - \theta_0) \sum x_i]$; since $\theta_1 < \theta_0$: $-(\theta_1 - \theta_0) > 0$; LR increasing in $\sum x_i$; reject for large $\sum X_i$

Q264. UMP One Sided Composite [HIGH]

For $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$ in exponential family with MLR in $T(x)$:

- **A)** UMP test rejects for $T(x) > c$ where c satisfies $\sup_{\theta \leq \theta_0} P_\theta(T > c) = \alpha$
- **B)** UMP test rejects for $T(x) < c$
- **C)** UMP does not exist for composite H_0
- **D)** UMP rejects for $|T(x)| > c$

Answer: A

Trap/Note: MLR property: test that is MP for each simple $\theta_1 > \theta_0$ is the same test; this common test is UMP; size achieved at $\theta = \theta_0$ (boundary)

Q265. Neyman Pearson Randomized [HIGH]

When $P_{H_0}(\Lambda > k) = \alpha_1 < \alpha < \alpha_2 = P_{H_0}(\Lambda \geq k)$, the NP test randomizes at $\Lambda = k$ with probability:

- **A)** $\gamma = (\alpha - \alpha_1) / (\alpha_2 - \alpha_1)$
- **B)** $\gamma = \alpha / \alpha_2$
- **C)** $\gamma = (\alpha_2 - \alpha) / (\alpha_2 - \alpha_1)$
- **D)** $\gamma = \alpha$

Answer: A

Trap/Note: Randomization probability γ chosen so total size = $\alpha_1 + \gamma(\alpha_2 - \alpha_1) = \alpha$; solving gives A; needed for discrete distributions

Q266. Power Function Monotone [HIGH]

For UMP test in exponential family with MLR, power function $\beta(\theta)$ is:

- **A)** Monotone increasing in θ (for one-sided $H_1 : \theta > \theta_0$)
- **B)** Monotone decreasing
- **C)** Non-monotone (U-shaped)

- D) Constant

Answer: A

Trap/Note: MLR implies monotone power function; this is the key property that makes the test UMP; power increases as θ moves further into H_1

Q267. Most Powerful Level [HIGH]

A test is most powerful of level α (not size α) if:

- A) $\sup_{\theta \in H_0} \beta(\theta) \leq \alpha$ and power is maximized
- B) $\beta(\theta_0) = \alpha$ exactly
- C) Power equals $1 - \beta$ for all $\theta \in H_1$
- D) It minimizes Type II error

Answer: A

Trap/Note: Level α : size $\leq \alpha$ (inequality); size α : size $= \alpha$ (equality); MP of level α may have size $< \alpha$ (conservative)

Q268. NP Lemma Uniform [HIGH]

For $X \sim U(0, \theta)$, testing $H_0 : \theta = 1$ vs $H_1 : \theta = 2$, MP test:

- A) Rejects if $X > c$ for some $c \in (0, 1)$
- B) Rejects if $X < c$
- C) Rejects if $X > 1$ (impossible under H_0)
- D) Always rejects (since H_1 support contains H_0 support)

Answer: A

Trap/Note: $f(x; 2)/f(x; 1) = 1/2$ for $x \in (0, 1)$ and ∞ for $x \in (1, 2)$; always reject if $X > 1$; for $X \leq 1$, LR = $1/2$ (constant); randomize to achieve exact size α

Q269. UMP Bernoulli [MEDIUM]

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, UMP test of $H_0 : p \leq p_0$ vs $H_1 : p > p_0$:

- A) Reject for large $\sum X_i$
- B) Reject for small $\sum X_i$
- C) Reject for $|\sum X_i - np_0| > c$
- D) UMP does not exist

Answer: A

Trap/Note: Bernoulli is exponential family with MLR in $\sum X_i$; UMP rejects for large $\sum X_i$; critical value from $\text{Bin}(n, p_0)$ distribution

Q270. Power Envelope [HIGH]

The power envelope is:

- A) Maximum achievable power at each $\theta \in H_1$ over all level- α tests
- B) Power of the UMP test

- C) Power of the likelihood ratio test
- D) Both A and B when UMP exists

Answer: D

Trap/Note: Power envelope = upper bound on power; when UMP exists, it achieves the envelope; when UMP doesn't exist, no single test achieves the envelope everywhere

Q271. NP Lemma Composite H1 [HIGH]

NP lemma directly applies to:

- A) Simple H_0 vs simple H_1 only
- B) Simple H_0 vs composite H_1
- C) Composite H_0 vs composite H_1
- D) Any hypothesis testing problem

Answer: A

Trap/Note: NP lemma: both hypotheses must be simple (point hypotheses); extension to composite uses UMP theory (MLR, exponential family); LRT handles general composite

Q272. Similar Test Construction [HIGH]

For composite $H_0 : \theta \in \Theta_0$, similar tests of size α are constructed using:

- A) Neyman structure: $E_\theta[\phi(X)|T(X)] = \alpha$ for sufficient statistic T under H_0
- B) Rao-Blackwell theorem
- C) Likelihood ratio
- D) Bootstrap

Answer: A

Trap/Note: Neyman structure: conditional on sufficient statistic for nuisance parameter, test has constant conditional size α ; this gives similar tests

Q273. UMP Unbiased Two Sided [HIGH]

UMPU test for $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$ in exponential family:

- A) Rejects for $T < c_1$ or $T > c_2$ with c_1, c_2 chosen for size and unbiasedness
- B) Rejects for $|T - \theta_0| > c$
- C) Is the same as the two-sided z -test always
- D) Does not exist

Answer: A

Trap/Note: UMPU: two critical values $c_1 < c_2$; conditions: $E_{\theta_0}[\phi] = \alpha$ (size) and $E_{\theta_0}[T\phi] = \alpha E_{\theta_0}[T]$ (unbiasedness); for normal, reduces to equal-tails z -test

Q274. Power Sample Size [MEDIUM]

To achieve power $1 - \beta$ at $\mu = \mu_1$ for one-sided z -test at level α , required n is:

- A) $n = \sigma^2(z_\alpha + z_\beta)^2 / (\mu_1 - \mu_0)^2$

- **B)** $n = \sigma^2(z_\alpha - z_\beta)^2/(\mu_1 - \mu_0)^2$
- **C)** $n = \sigma^2 z_\alpha^2/(\mu_1 - \mu_0)^2$
- **D)** $n = \sigma^2(z_{\alpha/2} + z_\beta)^2/(\mu_1 - \mu_0)^2$

Answer: A

Trap/Note: Power = $\Phi(\delta\sqrt{n}/\sigma - z_\alpha) = 1 - \beta$; so $\delta\sqrt{n}/\sigma = z_\alpha + z_\beta$; squaring: A; D uses $z_{\alpha/2}$ (two-sided)

Q275. NP Lemma Sufficient Stat [HIGH]

When sufficient statistic T exists, NP lemma critical region can be expressed in terms of T because:

- **A)** Likelihood ratio is a function of T only
- **B)** T has minimum variance
- **C)** T is complete
- **D)** T is ancillary

Answer: A

Trap/Note: By factorization theorem, $f(\mathbf{x}; \theta) = g(T; \theta)h(\mathbf{x})$; LR = $g(T; \theta_1)/g(T; \theta_0)$; depends on $\mathbf{x}$ only through T

Q276. Power Function Beta Distribution [HIGH]

For $X \sim \text{Beta}(\theta, 1)$, testing $H_0 : \theta = 1$ vs $H_1 : \theta = 2$, power of MP test at level α :

- **A)** $1 - (1 - \alpha)^2$
- **B)** α^2
- **C)** $2\alpha - \alpha^2$
- **D)** $1 - \alpha^2$

Answer: A

Trap/Note: $f(x; \theta) = \theta x^{\theta-1}$; LR = $2x/1 = 2x$; reject for large x ; $P(X > c | \theta = 1) = 1 - c = \alpha$; $c = 1 - \alpha$; power = $P(X > 1 - \alpha | \theta = 2) = 1 - (1 - \alpha)^2$

Q277. UMP Scale Family [HIGH]

For scale family $\frac{1}{\theta}f(x/\theta)$ with f log-concave, UMP test of $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$:

- **A)** Rejects for large $\sum X_i$ (if f is exponential)
- **B)** Rejects for large $X_{(n)}$ (if f is uniform)
- **C)** Depends on specific f
- **D)** Does not exist in general

Answer: C

Trap/Note: Sufficient statistic and rejection region depend on the specific scale family; exponential: $\sum X_i$; uniform: $X_{(n)}$; no universal answer

Q278. Critical Region Composite [HIGH]

For composite $H_0 : \mu \geq \mu_0$ vs $H_1 : \mu < \mu_0$ (normal, σ known), level- α test:

- A) Rejects for $\bar{X} < \mu_0 - z_\alpha \sigma / \sqrt{n}$
- B) Rejects for $\bar{X} > \mu_0 + z_\alpha \sigma / \sqrt{n}$
- C) Rejects for $|\bar{X} - \mu_0| > z_{\alpha/2} \sigma / \sqrt{n}$
- D) Rejects for $\bar{X} < \mu_0 - z_{\alpha/2} \sigma / \sqrt{n}$

Answer: A

Trap/Note: One-sided left test; size achieved at $\mu = \mu_0$ (boundary of H_0); use z_α not $z_{\alpha/2}$; B is right-sided; C is two-sided

Q279. Power Function Continuity [HIGH]

Power function $\beta(\theta)$ of a test based on continuous data is:

- A) Always continuous in θ
- B) Continuous only for exponential families
- C) May be discontinuous for discrete distributions
- D) Always differentiable

Answer: A

Trap/Note: For continuous data, power function is continuous (and often differentiable) in θ ; for discrete data, power function may have jumps; A is correct for continuous case

Q280. NP Lemma Multiparameter [HIGH]

For multiparameter family, NP lemma:

- A) Applies directly to simple vs simple hypotheses
- B) Does not apply
- C) Requires sufficient statistic
- D) Gives UMP test always

Answer: A

Trap/Note: NP lemma applies to any simple vs simple hypothesis regardless of dimension; the LR test is MP; extension to composite is harder in multiparameter case

Q281. UMP Exponential Rate [MEDIUM]

For $X_1, \dots, X_n \sim \text{Exp}(\theta)$ (rate), UMP test of $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$:

- A) Reject for small $\sum X_i$ (large rate = small mean)
- B) Reject for large $\sum X_i$
- C) Reject for large $\bar{X}$
- D) UMP does not exist

Answer: A

Trap/Note: Large rate θ means small mean $1/\theta$; LR increasing in $-\sum X_i$ (or decreasing in $\sum X_i$); reject for SMALL $\sum X_i$; common sign error

Q282. Power Function Normal Two Sided [HIGH]

Power function of two-sided z -test for $H_0 : \mu = 0$ vs $H_1 : \mu \neq 0$:

- **A)** $\beta(\mu) = \Phi(-z_{\alpha/2} - \mu\sqrt{n}/\sigma) + 1 - \Phi(z_{\alpha/2} - \mu\sqrt{n}/\sigma)$
- **B)** $\beta(\mu) = \Phi(\mu\sqrt{n}/\sigma - z_{\alpha/2})$
- **C)** $\beta(\mu) = 1 - \Phi(z_{\alpha/2} - \mu\sqrt{n}/\sigma)$
- **D)** $\beta(\mu) = 2\Phi(\mu\sqrt{n}/\sigma - z_{\alpha/2}) - 1$

Answer: A

Trap/Note: Two-sided: power = P(reject left) + P(reject right); for $\mu > 0$: left tail negligible; A is exact; B and C are one-sided approximations

Q283. Similar Region Exponential Family [HIGH]

In exponential family $f(x; \theta, \lambda) = h(x) \exp[\eta_1(\theta)T_1 + \eta_2(\lambda)T_2 - B]$, similar test for $H_0 : \theta = \theta_0$ (nuisance λ) conditions on:

- **A)** T_2 (sufficient for λ)
- **B)** T_1 (sufficient for θ)
- **C)** (T_1, T_2) jointly
- **D)** Neither

Answer: A

Trap/Note: Condition on sufficient statistic for nuisance parameter λ ; conditional distribution of $T_1|T_2$ is free of λ ; gives similar test

Q284. UMP Double Exponential [HIGH]

For $X_1, \dots, X_n \sim \text{Laplace}(\theta, 1)$, UMP test of $H_0 : \theta = 0$ vs $H_1 : \theta > 0$:

- **A)** Rejects for large $\sum X_i$
- **B)** Rejects for large $\sum |X_i|$
- **C)** Rejects for large sample median
- **D)** UMP does not exist

Answer: D

Trap/Note: Laplace is NOT exponential family (absolute value in exponent); no MLR in simple statistic; UMP does not exist; sign test or Wilcoxon are used

Q285. Power Function Beta Calculation [HIGH]

For $X \sim \text{Bin}(10, p)$, test $H_0 : p = 0.5$ vs $H_1 : p > 0.5$, reject if $X \geq 8$. Power at $p = 0.7$:

- **A)** $P(X \geq 8|p = 0.7) = \binom{10}{8}(0.7)^8(0.3)^2 + \binom{10}{9}(0.7)^9(0.3) + \binom{10}{10}(0.7)^{10}$
- **B)** $P(X \geq 8|p = 0.5)$
- **C)** $1 - P(X \leq 7|p = 0.7)$
- **D)** Both A and C (equivalent)

Answer: D

Trap/Note: Power at $p = 0.7$: compute under H_1 ; A and C are equivalent expressions; B is the size (under H_0)

Q286. NP Lemma Gamma [HIGH]

For $X_1, \dots, X_n \sim \Gamma(\alpha, \theta)$ with α known, MP test of $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1 < \theta_0$:

- A) Reject for large $\sum X_i$
- B) Reject for small $\sum X_i$
- C) Reject for large $\bar{X}$
- D) Both A and C (equivalent)

Answer: B

Trap/Note: $\Gamma(\alpha, \theta)$ with scale θ : $f \propto \theta^{-n\alpha} \exp(-\sum x_i/\theta)$; LR = $(\theta_0/\theta_1)^{n\alpha} \exp[-\sum x_i(1/\theta_1 - 1/\theta_0)]$; since $\theta_1 < \theta_0$: $1/\theta_1 > 1/\theta_0$; LR decreasing in $\sum x_i$; reject for SMALL $\sum X_i$

Q287. UMP Normal Mean Variance Unknown [HIGH]

For $N(\mu, \sigma^2)$ with σ^2 unknown, UMP test of $H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$:

- A) Rejects for large $t = (\bar{X} - \mu_0)/(S/\sqrt{n})$
- B) Rejects for large $\bar{X}$
- C) UMP does not exist (nuisance σ^2)
- D) Rejects for large $\bar{X}/S$

Answer: A

Trap/Note: With nuisance σ^2 , UMP doesn't exist in general; BUT the t -test is UMPU (uniformly most powerful unbiased); A is UMPU, not UMP; C is technically correct for UMP

Q288. Power Function Increasing [HIGH]

For one-sided test in exponential family, power function is strictly increasing because:

- A) MLR property implies power is monotone in θ
- B) Sufficient statistic is complete
- C) MLE is consistent
- D) CRLB is attained

Answer: A

Trap/Note: MLR $\Rightarrow$ monotone power function; this is the fundamental connection between MLR and UMP tests; completeness is separate

Q289. NP Lemma Composite Reduction [HIGH]

For composite $H_1 : \theta > \theta_0$ in exponential family, UMP test is found by:

- A) Applying NP lemma to each simple $H_1 : \theta = \theta_1$ and noting the MP test is the same for all $\theta_1 > \theta_0$
- B) Maximizing power at $\theta_1 = \theta_0 + 1$
- C) Using LRT
- D) Rao-Blackwell theorem

Answer: A

Trap/Note: Key insight: MLR ensures the NP critical region (reject for large T) is the same for ALL $\theta_1 > \theta_0$; this common test is UMP

Q290. Critical Region Randomized Discrete [HIGH]

For $X \sim \text{Bin}(5, p)$, testing $H_0 : p = 0.5$ at level $\alpha = 0.05$, exact size requires:

- A) Randomization at boundary since $P(X \geq 5 | p = 0.5) = 1/32 < 0.05 < P(X \geq 4 | p = 0.5) = 6/32$
- B) Reject if $X \geq 5$ (size = $1/32 < 0.05$, conservative)
- C) Reject if $X \geq 4$ (size = $6/32 > 0.05$, liberal)
- D) Use normal approximation

Answer: A

Trap/Note: Exact size 0.05 requires randomization: reject if $X = 5$; reject with prob γ if $X = 4$; $\gamma = (0.05 - 1/32)/(6/32 - 1/32) = (0.05 - 0.03125)/0.15625$

Q291. UMP Negative Binomial [HIGH]

For $X_1, \dots, X_n \sim \text{NegBin}(r, p)$ with r known, UMP test of $H_0 : p \geq p_0$ vs $H_1 : p < p_0$:

- A) Reject for large $\sum X_i$
- B) Reject for small $\sum X_i$
- C) Reject for large $\bar{X}$
- D) Both A and C (equivalent)

Answer: D

Trap/Note: NegBin: $E[X] = r(1-p)/p$; smaller p means larger mean; LR in $\sum X_i$; for $p_1 < p_0$: reject for large $\sum X_i$; A and C equivalent

Q292. Power Function Envelope Calculation [HIGH]

Power envelope at θ_1 for testing $H_0 : \theta = \theta_0$ at level α is:

- A) Power of NP MP test at θ_1
- B) $1 - \beta$ where β is Type II error of best test
- C) $\sup_{\text{level-}\alpha \text{ tests}} \beta(\theta_1)$
- D) All of the above (equivalent)

Answer: D

Trap/Note: All three definitions are equivalent; NP lemma gives the test achieving the envelope; power envelope is the theoretical maximum

Q293. Similar Region Normal [HIGH]

For $N(\mu, \sigma^2)$ with both unknown, similar test of $H_0 : \mu = \mu_0$ conditions on:

- A) S^2 (sufficient for σ^2 under H_0)
- B) $\bar{X}$
- C) $\sum X_i^2$
- D) $\bar{X}_{(1)}$

Answer: A

Trap/Note: Under $H_0 : \mu = \mu_0$, S^2 is sufficient for σ^2 ; conditioning on S^2 gives t -statistic with t_{n-1} distribution free of σ^2 ; this gives the t -test as UMPU

Q294. UMP Pareto [HIGH]

For $X_1, \dots, X_n \sim \text{Pareto}(\theta)$, $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, UMP test of $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$:

- A) Reject for large $\sum \log X_i$
- B) Reject for small $\sum \log X_i$
- C) Reject for large $\prod X_i$
- D) Both B and C (equivalent since $\log \prod = \sum \log$)

Answer: B

Trap/Note: $\sum \log X_i \sim \Gamma(n, 1/\theta)$; larger θ means smaller $\sum \log X_i$; MLR in $-\sum \log X_i$; reject for small $\sum \log X_i$ (= large $-\sum \log X_i$)

Q295. Power Function Beta Distribution General [HIGH]

For $X \sim \text{Beta}(\theta, 1)$, UMP test of $H_0 : \theta \leq 1$ vs $H_1 : \theta > 1$ rejects for:

- A) Large $\sum \log X_i$ (less negative, closer to 0)
- B) Small $\sum \log X_i$ (more negative)
- C) Large $\sum X_i$
- D) Large $X_{(n)}$

Answer: A

Trap/Note: $\log X_i < 0$ for $X_i \in (0, 1)$; $-\log X_i \sim \text{Exp}(\theta)$; large θ means small $E[-\log X_i] = 1/\theta$; so large θ means $\sum \log X_i$ closer to 0 (less negative); reject for large (less negative) $\sum \log X_i$

Q296. NP Lemma Location Family [HIGH]

For location family $f(x - \theta)$ with f symmetric and log-concave, MP test of $H_0 : \theta = 0$ vs $H_1 : \theta = \delta > 0$:

- A) Rejects for large $\sum X_i$ (if f is normal)
- B) Rejects for large $\sum \text{sgn}(X_i)$ (if f is Laplace)
- C) Depends on specific f
- D) Always rejects for large $\bar{X}$

Answer: C

Trap/Note: MP test depends on f ; normal: reject for large $\bar{X}$; Laplace: reject for large $\sum \text{sgn}(X_i)$ (sign test); Cauchy: complex; D is wrong in general

Q297. UMP Hypergeometric [HIGH]

For Hypergeometric(N, M, n) with N, n known, UMP test of $H_0 : M \leq M_0$ vs $H_1 : M > M_0$:

- A) Reject for large X (number of successes in sample)
- B) Reject for small X
- C) UMP does not exist (not exponential family)
- D) Reject for $X > M_0 n / N$

Answer: A

Trap/Note: Hypergeometric has MLR in X ; UMP exists and rejects for large X ; not exponential family but MLR still holds

Q298. Power Function Composite H0 [HIGH]

For composite $H_0 : \theta \in [0, 1]$ vs $H_1 : \theta > 1$, size of test is:

- **A)** $\sup_{\theta \in [0, 1]} \beta(\theta)$
- **B)** $\beta(0)$
- **C)** $\beta(1)$
- **D)** $\inf_{\theta \in [0, 1]} \beta(\theta)$

Answer: A

Trap/Note: Size = supremum of power function over H_0 ; for monotone power function, supremum achieved at boundary $\theta = 1$; so $A = \beta(1)$ in this case

Q299. NP Lemma Sufficient Reduction [HIGH]

If T is sufficient and NP critical region is $\{x : \Lambda(x) > k\}$, then equivalently:

- **A)** $\{x : g(T(x)) > k'\}$ for some function g and constant k'
- **B)** $\{x : T(x) > k'\}$ always
- **C)** $\{x : \bar{X} > k'\}$ always
- **D)** The critical region cannot be simplified

Answer: A

Trap/Note: LR is a function of T (by factorization); so critical region is a function of T ; B is true only if LR is monotone in T ; A is the general statement

Q310. UMP Summary [HIGH]

UMP test exists for $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$ (two-sided) when:

- **A)** Distribution is normal
- **B)** Distribution has MLR
- **C)** Never (for two-sided alternatives in general)
- **D)** Distribution is from exponential family

Answer: C

Trap/Note: UMP for two-sided alternatives generally does NOT exist; UMPU exists for exponential families; this is a fundamental limitation of UMP theory

Q311–Q365. Likelihood Ratio Tests, SPRT & Decision Theory

Q311. LRT Definition [MEDIUM]

Likelihood ratio test statistic Λ is:

- A) $\Lambda = \sup_{\theta \in \Theta_0} L(\theta) / \sup_{\theta \in \Theta} L(\theta)$
- B) $\Lambda = \sup_{\theta \in \Theta} L(\theta) / \sup_{\theta \in \Theta_0} L(\theta)$
- C) $\Lambda = L(\hat{\theta}_0) / L(\hat{\theta})$ where $\hat{\theta}_0$ is restricted MLE
- D) Both A and C (equivalent)

Answer: D

Trap/Note: $\Lambda \in [0, 1]$; reject for SMALL Λ (equivalently large $-2 \log \Lambda$); B is ≥ 1 always; A and C are equivalent

Q312. LRT Wilks Theorem [MEDIUM]

Under H_0 and regularity conditions, $-2 \log \Lambda \xrightarrow{d}$:

- A) χ_r^2 where $r = \dim(\Theta) - \dim(\Theta_0)$
- B) χ_n^2
- C) $F_{r, n-r}$
- D) $N(0, 1)$

Answer: A

Trap/Note: Wilks' theorem: r = number of restrictions (difference in parameter dimensions); for testing k parameters: $r = k$; requires regularity conditions

Q313. LRT Normal Mean Variance [HIGH]

For $N(\mu, \sigma^2)$ with both unknown, LRT of $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$:

- A) Reduces to t -test: reject for large $|t| = |\bar{X} - \mu_0| / (S / \sqrt{n})$
- B) Reduces to z -test
- C) Reduces to χ^2 test
- D) Has no closed form

Answer: A

Trap/Note: LRT statistic: $-2 \log \Lambda = n \log[1 + t^2 / (n - 1)]$ which is monotone in $|t|$; LRT is equivalent to t -test; $r = 1$ df

Q314. SPRT Definition [MEDIUM]

Wald's SPRT continues sampling while:

- A) $B < \Lambda_m < A$ where $\Lambda_m = \prod_{i=1}^m f(X_i; \theta_1) / f(X_i; \theta_0)$
- B) $\Lambda_m > A$
- C) $\Lambda_m < B$
- D) $m < n_0$ (fixed sample size)

Answer: A

Trap/Note: SPRT: continue if $B < \Lambda_m < A$; stop and reject H_0 if $\Lambda_m \geq A$; stop and accept H_0 if $\Lambda_m \leq B$

Q315. SPRT Boundaries [MEDIUM]

Wald's approximate SPRT boundaries are:

- **A)** $A \approx (1 - \beta)/\alpha$, $B \approx \beta/(1 - \alpha)$
- **B)** $A \approx (1 - \alpha)/\beta$, $B \approx \alpha/(1 - \beta)$
- **C)** $A = 1/\alpha$, $B = \beta$
- **D)** $A = 1 - \beta$, $B = \alpha$

Answer: A

Trap/Note: $A = (1 - \beta)/\alpha$ (upper boundary for rejecting H_0); $B = \beta/(1 - \alpha)$ (lower boundary for accepting H_0); these are approximate (ignore overshoot)

Q316. SPRT Normal [MEDIUM]

For $X_i \sim N(\theta, 1)$, SPRT of $H_0 : \theta = 0$ vs $H_1 : \theta = \delta$, log-LR after m observations:

- **A)** $\delta \sum_{i=1}^m X_i - m\delta^2/2$
- **B)** $\delta \sum_{i=1}^m X_i$
- **C)** $\delta \bar{X}_m - \delta^2/2$
- **D)** $\sum_{i=1}^m X_i^2$

Answer: A

Trap/Note: $\log \Lambda_m = \sum [\log f(X_i; \delta) - \log f(X_i; 0)] = \delta \sum X_i - m\delta^2/2$; linear in $\sum X_i$; SPRT reduces to boundary on $\sum X_i$

Q317. OC Function Definition [MEDIUM]

Operating Characteristic (OC) function $L(\theta)$ of SPRT is:

- **A)** $P_\theta(\text{accept } H_0)$
- **B)** $P_\theta(\text{reject } H_0)$
- **C)** $E_\theta[N]$ (expected sample size)
- **D)** Power function

Answer: A

Trap/Note: OC function = probability of accepting H_0 ; $L(\theta_0) = 1 - \alpha$; $L(\theta_1) = \beta$; power function = $1 - L(\theta)$; C is ASN function

Q318. ASN Function [MEDIUM]

ASN (Average Sample Number) function $E_\theta[N]$ for SPRT satisfies Wald's identity:

- **A)** $E_\theta[N] \approx \frac{L(\theta) \log(B/A) + (1 - L(\theta)) \log(A/B)}{E_\theta[\log f(X; \theta_1)/f(X; \theta_0)]}$ (Wald's approximation)
- **B)** $E_\theta[N] = n$ (fixed)
- **C)** $E_\theta[N] = 1/\alpha$
- **D)** $E_\theta[N] = \log(A/B)/E_\theta[\log \Lambda_1]$

Answer: A

Trap/Note: Wald's approximation for ASN uses OC function $L(\theta)$ and expected log-LR per observation; D is a special case; exact formula involves overshoot correction

Q319. SPRT Optimality [HIGH]

Wald-Wolfowitz theorem states SPRT is optimal in that:

- **A)** Among all tests with error probabilities $\leq \alpha$ and $\leq \beta$, SPRT minimizes $E_{\theta_0}[N]$ and $E_{\theta_1}[N]$ simultaneously
- **B)** SPRT minimizes maximum expected sample size
- **C)** SPRT is UMP among sequential tests
- **D)** SPRT minimizes total error probability

Answer: A

Trap/Note: Wald-Wolfowitz: SPRT minimizes ASN at BOTH θ_0 and θ_1 simultaneously; no other test with same error bounds can have smaller ASN at either point

Q320. LRT Composite Both [HIGH]

LRT for $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1 = \Theta \setminus \Theta_0$:

- **A)** $\Lambda = \sup_{\Theta_0} L / \sup_{\Theta} L$; reject for small Λ
- **B)** $\Lambda = \sup_{\Theta_1} L / \sup_{\Theta_0} L$; reject for large Λ
- **C)** $\Lambda = L(\hat{\theta}_0) / L(\hat{\theta})$; reject for small Λ
- **D)** Both A and C (equivalent)

Answer: D

Trap/Note: A and C are equivalent; $\sup_{\Theta} L = \max(\sup_{\Theta_0} L, \sup_{\Theta_1} L)$; when $\hat{\theta} \notin \Theta_0$, $\sup_{\Theta} L = L(\hat{\theta})$; reject for small Λ (large $-2 \log \Lambda$)

Q321. Decision Theory Loss [MEDIUM]

In decision theory, risk function $R(\theta, d)$ is:

- **A)** $E_{\theta}[L(\theta, d(X))]$ — expected loss
- **B)** $L(\theta, d(X))$ — loss function
- **C)** $\sup_{\theta} L(\theta, d(X))$
- **D)** $P_{\theta}(\text{wrong decision})$

Answer: A

Trap/Note: Risk = expected loss; loss function is random (depends on X); risk is its expectation; D is risk for 0-1 loss only

Q322. Decision Theory Admissibility [HIGH]

Estimator d_1 dominates d_2 if:

- **A)** $R(\theta, d_1) \leq R(\theta, d_2)$ for all θ with strict inequality for some θ
- **B)** $R(\theta, d_1) < R(\theta, d_2)$ for all θ

- C) $R(\theta, d_1) \leq R(\theta, d_2)$ for all θ
- D) $\sup_{\theta} R(\theta, d_1) < \sup_{\theta} R(\theta, d_2)$

Answer: A

Trap/Note: Dominance: $\leq$ everywhere AND $<$ somewhere; B is too strong (strict everywhere); C allows equality everywhere (no dominance); D is minimax comparison

Q323. Minimax Estimator [HIGH]

Minimax estimator minimizes:

- A) $\sup_{\theta} R(\theta, d)$ — worst-case risk
- B) $\int R(\theta, d) \pi(\theta) d\theta$ — Bayes risk
- C) $R(\theta_0, d)$ at a specific θ_0
- D) $E_{\theta}[d(X)]$ — expected estimate

Answer: A

Trap/Note: Minimax: minimize maximum risk; Bayes: minimize average risk (B); minimax estimator is often Bayes w.r.t. least favorable prior

Q324. Bayes Estimator [MEDIUM]

Bayes estimator under squared error loss is:

- A) Posterior mean $E[\theta|X]$
- B) Posterior median
- C) Posterior mode (MAP)
- D) Prior mean

Answer: A

Trap/Note: Squared error: posterior mean; absolute error: posterior median; 0-1 loss: posterior mode; these are fundamental results

Q325. SPRT Poisson [HIGH]

For $X_i \sim \text{Poisson}(\lambda)$, SPRT of $H_0 : \lambda = \lambda_0$ vs $H_1 : \lambda = \lambda_1 > \lambda_0$, log-LR:

- A) $\sum X_i \log(\lambda_1/\lambda_0) - m(\lambda_1 - \lambda_0)$
- B) $\sum X_i \log(\lambda_1/\lambda_0)$
- C) $m(\lambda_1 - \lambda_0)$
- D) $\bar{X} \log(\lambda_1/\lambda_0)$

Answer: A

Trap/Note: $\log \Lambda_m = \sum [x_i \log(\lambda_1/\lambda_0) - (\lambda_1 - \lambda_0)] = \sum X_i \log(\lambda_1/\lambda_0) - m(\lambda_1 - \lambda_0)$; linear in $\sum X_i$; SPRT boundary on $\sum X_i$

Q326. LRT Independence [HIGH]

LRT for independence in $r \times c$ contingency table: $-2 \log \Lambda \xrightarrow{d} \chi_{(r-1)(c-1)}^2$ because:

- A) $\dim(\Theta) = rc - 1$, $\dim(\Theta_0) = (r - 1) + (c - 1)$; $r = (rc - 1) - (r + c - 2) = (r - 1)(c - 1)$

- B) Degrees of freedom = rc
- C) Degrees of freedom = $r + c - 2$
- D) Wilks theorem gives χ^2_1 always

Answer: A

Trap/Note: Saturated model: $rc - 1$ free parameters; independence model: $(r - 1) + (c - 1)$ free parameters; difference = $(r - 1)(c - 1)$; this is the df for χ^2

Q327. Decision Theory Bayes Risk [HIGH]

Bayes risk of estimator d w.r.t. prior π is:

- A) $r(\pi, d) = \int R(\theta, d)\pi(\theta)d\theta = E_\pi[R(\theta, d)]$
- B) $r(\pi, d) = \sup_\theta R(\theta, d)$
- C) $r(\pi, d) = R(\theta_0, d)$ at prior mean
- D) $r(\pi, d) = E[L(\theta, d(X))]$ over joint distribution

Answer: D

Trap/Note: A and D are equivalent (both are expectation of loss over joint distribution of (θ, X)); B is minimax risk; Bayes estimator minimizes A=D

Q328. SPRT Exponential [HIGH]

For $X_i \sim \text{Exp}(\theta)$ (rate), SPRT of $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1 > \theta_0$, SPRT boundary on $\sum X_i$:

- A) Accept H_0 if $\sum X_i > b$, reject if $\sum X_i < a$ (for some $a < b$)
- B) Accept H_0 if $\sum X_i < a$, reject if $\sum X_i > b$
- C) Reject for large $\sum X_i$
- D) Reject for small $\bar{X}$

Answer: A

Trap/Note: Large θ (rate) means small mean; log-LR = $n \log(\theta_1/\theta_0) - (\theta_1 - \theta_0) \sum X_i$; decreasing in $\sum X_i$; large $\sum X_i$ favors H_0 (small rate); accept H_0 for large $\sum X_i$

Q329. LRT Normal Variance [HIGH]

LRT for $H_0 : \sigma^2 = \sigma_0^2$ vs $H_1 : \sigma^2 \neq \sigma_0^2$ (normal, μ unknown):

- A) Reduces to χ^2 test: reject for large or small $(n - 1)S^2/\sigma_0^2$
- B) Reduces to F -test
- C) Reduces to z -test
- D) Has no closed form

Answer: A

Trap/Note: LRT statistic: $-2 \log \Lambda = n[\log(\hat{\sigma}^2/\sigma_0^2) + \sigma_0^2/\hat{\sigma}^2 - 1]$ which is monotone function of $(n - 1)S^2/\sigma_0^2$; equivalent to χ^2 test

Q330. OC Function SPRT Normal [HIGH]

For SPRT of $H_0 : \theta = 0$ vs $H_1 : \theta = \delta$ with $X_i \sim N(\theta, 1)$, OC function $L(\theta)$ satisfies:

- A) $L(\theta) = \frac{A^{h(\theta)} - 1}{A^{h(\theta)} - B^{h(\theta)}}$ where $h(\theta)$ solves $E_\theta[e^{h \log \Lambda_1}] = 1$
- B) $L(\theta) = 1 - \Phi(\theta/\sqrt{n})$
- C) $L(\theta) = \Phi(-\theta/\sqrt{n})$
- D) $L(\theta) = \beta$ for all $\theta \in H_1$

Answer: A

Trap/Note: Wald's OC approximation uses moment generating function of log-LR; $h(\theta)$ is the non-trivial root of $E_\theta[\Lambda_1^h] = 1$; complex but standard formula

Q331. Decision Theory Complete Class [HIGH]

Complete class theorem states:

- A) Every admissible estimator is Bayes (or limit of Bayes)
- B) Every Bayes estimator is admissible
- C) Both A and B
- D) Admissible estimators form a complete class

Answer: C

Trap/Note: B: Bayes estimators with proper priors are admissible; A: admissible estimators are Bayes (or generalized Bayes); together they characterize admissibility via Bayes

Q332. LRT Regression [HIGH]

LRT for $H_0 : \beta_1 = \dots = \beta_k = 0$ in multiple regression (normal errors):

- A) Reduces to F -test: $F = (\text{RSS}_0 - \text{RSS}_1)/(k\hat{\sigma}^2)$
- B) Reduces to t -test
- C) Reduces to χ_k^2 test
- D) Has no closed form

Answer: A

Trap/Note: $\text{LRT} = -2 \log \Lambda = n \log(\text{RSS}_0/\text{RSS}_1)$; for normal errors, this is monotone in F -statistic; LRT and F -test are equivalent

Q333. SPRT Bernoulli [MEDIUM]

For $X_i \sim \text{Bernoulli}(p)$, SPRT of $H_0 : p = p_0$ vs $H_1 : p = p_1 > p_0$, log-LR:

- A) $\sum X_i \log \frac{p_1(1-p_0)}{p_0(1-p_1)} + m \log \frac{1-p_1}{1-p_0}$
- B) $\sum X_i \log(p_1/p_0)$
- C) $m \log[(1-p_1)/(1-p_0)]$
- D) $\bar{X} \log(p_1/p_0) + \log[(1-p_1)/(1-p_0)]$

Answer: A

Trap/Note: $\log \Lambda_m = \sum x_i \log \frac{p_1}{p_0} + \sum (1-x_i) \log \frac{1-p_1}{1-p_0} = \sum x_i \log \frac{p_1(1-p_0)}{p_0(1-p_1)} + m \log \frac{1-p_1}{1-p_0}$; linear in $\sum X_i$

Q334. Minimax Normal Mean [HIGH]

For $X \sim N(\theta, 1)$, minimax estimator of θ under squared error loss is:

- A) $\bar{X}$ (sample mean)
- B) 0 (constant)
- C) $\bar{X}$ (since it is admissible and has constant risk)
- D) James-Stein estimator

Answer: C

Trap/Note: For $p = 1$: $\bar{X}$ is minimax (constant risk = $1/n$, admissible); for $p \geq 3$: James-Stein dominates $\bar{X}$ in MSE but $\bar{X}$ is still minimax; D is wrong for $p = 1$

Q335. LRT Goodness Of Fit [HIGH]

LRT for goodness of fit (multinomial): $G^2 = -2 \log \Lambda = 2 \sum O_i \log(O_i/E_i) \xrightarrow{d}$:

- A) χ^2_{k-1-p} where k = cells, p = estimated parameters
- B) χ^2_k
- C) χ^2_{k-1}
- D) $F_{k-1, n-k}$

Answer: A

Trap/Note: G^2 (log-likelihood ratio) and Pearson χ^2 are asymptotically equivalent; df = $k-1-p$ (subtract 1 for total constraint and p for estimated parameters)

Q336. Decision Theory Minimax Bayes [HIGH]

Minimax estimator equals Bayes estimator w.r.t. least favorable prior π^* when:

- A) Bayes risk equals minimax risk: $r(\pi^*, d^*) = \sup_{\theta} R(\theta, d^*)$
- B) Prior is uniform
- C) Loss function is squared error
- D) Sample size is large

Answer: A

Trap/Note: Least favorable prior: prior that maximizes Bayes risk; if Bayes estimator has constant risk, it is minimax; key connection between Bayes and minimax

Q337. SPRT ASN Comparison [HIGH]

SPRT has smaller ASN than fixed sample test because:

- A) It stops early when evidence is strong (in either direction)
- B) It uses more information per observation
- C) It has smaller error probabilities
- D) It uses sufficient statistics

Answer: A

Trap/Note: SPRT's efficiency comes from early stopping; when data strongly favors one hypothesis, no need to continue; fixed sample test always uses n observations

Q338. LRT Exponential Family [HIGH]

For exponential family, LRT of $H_0 : \eta = \eta_0$ vs $H_1 : \eta \neq \eta_0$:

- **A)** $-2 \log \Lambda = 2n[B(\hat{\eta}) - B(\eta_0) - (\hat{\eta} - \eta_0)\nabla B(\eta_0)]$ where B is log-partition
- **B)** $-2 \log \Lambda = n(\hat{\eta} - \eta_0)^2 I(\eta_0)$
- **C)** $-2 \log \Lambda = 2 \sum \log f(X_i; \hat{\eta}) / f(X_i; \eta_0)$
- **D)** All equivalent asymptotically

Answer: D

Trap/Note: All three are asymptotically equivalent (Wald, LRT, Score); A is exact LRT; B is Wald; they differ in finite samples

Q339. Decision Theory Regret [HIGH]

Regret of decision d at θ is:

- **A)** $R(\theta, d) - \inf_{d'} R(\theta, d')$ — excess risk over best possible
- **B)** $R(\theta, d)$ — total risk
- **C)** $\sup_{\theta} R(\theta, d) - R(\theta, d)$
- **D)** $E[L(\theta, d(X))]$

Answer: A

Trap/Note: Regret = how much worse than oracle (who knows θ); minimax regret minimizes worst-case regret; different from minimax risk

Q340. LRT Two Sample Normal [HIGH]

LRT for $H_0 : \mu_1 = \mu_2$ (two independent normal samples, equal σ^2 unknown):

- **A)** Reduces to two-sample t -test
- **B)** Reduces to F -test
- **C)** Reduces to z -test
- **D)** Has no closed form

Answer: A

Trap/Note: LRT statistic is monotone in t^2 ; equivalent to two-sample t -test with pooled variance; $r = 1$ df for Wilks

Q341. SPRT Geometric [HIGH]

For $X_i \sim \text{Geometric}(p)$, SPRT of $H_0 : p = p_0$ vs $H_1 : p = p_1 < p_0$, log-LR:

- **A)** $\sum X_i \log \frac{1-p_1}{1-p_0} + m \log \frac{p_1}{p_0}$
- **B)** $\sum X_i \log(p_1/p_0)$
- **C)** $m \log(p_1/p_0)$
- **D)** $\bar{X} \log[(1-p_1)/(1-p_0)]$

Answer: A

Trap/Note: $\log f(x; p) = \log p + (x-1) \log(1-p)$; $\log \Lambda_m = m \log(p_1/p_0) + \sum (X_i - 1) \log \frac{1-p_1}{1-p_0} = \sum X_i \log \frac{1-p_1}{1-p_0} + m \log \frac{p_1}{p_0}$

Q342. Decision Theory Stein Admissibility [HIGH]

James-Stein estimator $\hat{\theta}_{JS} = (1 - \frac{p-2}{n\|X\|^2})\bar{X}$ for $p \geq 3$:

- A) Dominates $\bar{X}$ in total MSE but is inadmissible itself
- B) Is admissible
- C) Dominates $\bar{X}$ and is admissible
- D) Is minimax

Answer: A

Trap/Note: James-Stein dominates $\bar{X}$ for $p \geq 3$ (Stein paradox); but JS itself is inadmissible — positive-part JS $\hat{\theta}_{JS}^+ = \max(0, 1 - \frac{p-2}{n\|X\|^2})\bar{X}$ dominates JS

Q343. LRT Homogeneity [HIGH]

LRT for homogeneity of k binomial proportions $H_0 : p_1 = \dots = p_k$:

- A) $G^2 = 2 \sum_{i,j} O_{ij} \log(O_{ij}/E_{ij}) \xrightarrow{d} \chi_{k-1}^2$
- B) $G^2 \xrightarrow{d} \chi_k^2$
- C) $G^2 \xrightarrow{d} \chi_{2k-2}^2$
- D) $G^2 \xrightarrow{d} F_{k-1, n-k}$

Answer: A

Trap/Note: k proportions, 1 constraint (common p): $r = k - 1$ df; same as Pearson χ^2 for $2 \times k$ table

Q344. SPRT Truncated [HIGH]

Truncated SPRT (with maximum sample size $n_{\max}$):

- A) Stops at $n_{\max}$ and makes decision based on accumulated evidence
- B) Is equivalent to fixed sample test
- C) Has same error probabilities as SPRT
- D) Is always more efficient than SPRT

Answer: A

Trap/Note: Truncated SPRT: if boundaries not crossed by $n_{\max}$, make decision at $n_{\max}$; error probabilities change slightly; practical compromise between SPRT and fixed sample

Q345. Decision Theory Posterior Risk [HIGH]

Posterior expected loss (posterior risk) of decision d given $X = x$ is:

- A) $\rho(\pi, d|x) = \int L(\theta, d)\pi(\theta|x)d\theta$
- B) $\rho(\pi, d|x) = \int L(\theta, d)\pi(\theta)d\theta$
- C) $\rho(\pi, d|x) = R(\theta, d)$
- D) $\rho(\pi, d|x) = E[L(\theta, d)|X = x]$

Answer: D

Trap/Note: A and D are equivalent (posterior expectation of loss); B uses prior not posterior; Bayes estimator minimizes posterior risk for each x

Q346. LRT Nested Models [HIGH]

For nested models $M_0 \subset M_1$, LRT statistic $-2 \log \Lambda$:

- **A)** Is always non-negative
- **B)** Equals $2[\ell(\hat{\theta}_1) - \ell(\hat{\theta}_0)]$ where ℓ is log-likelihood
- **C)** Has χ^2 distribution under M_0
- **D)** All of the above

Answer: D

Trap/Note: All three: A (log-likelihood increases with more parameters); B (definition); C (Wilks theorem); all hold under regularity

Q347. SPRT Composite [HIGH]

For composite hypotheses $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta \geq \theta_1$ ($\theta_0 < \theta_1$), SPRT:

- **A)** Uses $\Lambda_m = \prod f(X_i; \theta_1)/f(X_i; \theta_0)$ (simple vs simple at boundaries)
- **B)** Uses generalized likelihood ratio
- **C)** Does not exist
- **D)** Uses MLE at each stage

Answer: A

Trap/Note: For composite hypotheses with indifference zone (θ_0, θ_1) , SPRT uses simple hypotheses at boundaries; error probabilities guaranteed for all $\theta \leq \theta_0$ and $\theta \geq \theta_1$

Q348. Decision Theory Invariant [HIGH]

Invariant decision rules under group G of transformations satisfy:

- **A)** $d(gx) = g'd(\theta)$ for all $g \in G$ (equivariance)
- **B)** $d(gx) = d(x)$ for all $g \in G$
- **C)** Risk is constant over orbits of G
- **D)** Both A and C

Answer: D

Trap/Note: Equivariant estimators have constant risk over orbits; best equivariant estimator minimizes risk within equivariant class; Pitman estimator is best location equivariant

Q349. LRT Bartlett Correction [HIGH]

Bartlett correction for LRT improves χ^2 approximation by:

- **A)** Dividing $-2 \log \Lambda$ by $1 + b/n$ where b is Bartlett factor
- **B)** Adding b/n to $-2 \log \Lambda$
- **C)** Using χ^2_{r+1} instead of χ^2_r
- **D)** Multiplying by $n/(n-r)$

Answer: A

Trap/Note: Bartlett: $(-2 \log \Lambda)/(1 + b/n) \sim \chi^2_r$ to order $O(n^{-2})$; improves from $O(n^{-1})$ error; b computed from cumulants of log-likelihood

Q350. SPRT Efficiency [HIGH]

SPRT efficiency relative to fixed sample test (same error probabilities) is measured by:

- A) $n/E_\theta[N]$ — ratio of fixed to expected sequential sample size
- B) $E_\theta[N]/n$
- C) α/β
- D) Power ratio

Answer: A

Trap/Note: Efficiency > 1 means SPRT uses fewer observations on average; for normal testing, SPRT efficiency $\approx 50\%$ savings at H_0 and H_1 ; efficiency < 1 would mean SPRT is worse

Q351. Decision Theory Loss Functions [MEDIUM]

Under 0-1 loss $L(\theta, d) = I(\theta \neq d)$, Bayes estimator is:

- A) Posterior mode (MAP)
- B) Posterior mean
- C) Posterior median
- D) Prior mode

Answer: A

Trap/Note: 0-1 loss: minimize $P(\theta \neq d|X) = 1 - P(\theta = d|X)$; maximize $P(\theta = d|X)$; this is the posterior mode; for continuous θ , 0-1 loss requires careful definition

Q352. LRT Multivariate Normal [HIGH]

LRT for $H_0 : \mu = \mu_0$ in $N_p(\mu, \Sigma)$ with Σ known:

- A) $-2 \log \Lambda = n(\bar{X} - \mu_0)^T \Sigma^{-1}(\bar{X} - \mu_0) \sim \chi_p^2$
- B) $-2 \log \Lambda = (\bar{X} - \mu_0)^T \Sigma^{-1}(\bar{X} - \mu_0) \sim \chi_p^2$
- C) $-2 \log \Lambda \sim F_{p, n-p}$
- D) $-2 \log \Lambda = n(\bar{X} - \mu_0)^T S^{-1}(\bar{X} - \mu_0) \sim \chi_p^2$

Answer: A

Trap/Note: $n(\bar{X} - \mu_0)^T \Sigma^{-1}(\bar{X} - \mu_0) = T^2$ (Hotelling's T^2 when Σ known); $\sim \chi_p^2$; D uses S (unknown Σ case) giving $T^2 \sim \frac{p(n-1)}{n-p} F_{p, n-p}$

Q353. SPRT Normal Variance [HIGH]

For $X_i \sim N(0, \sigma^2)$, SPRT of $H_0 : \sigma^2 = \sigma_0^2$ vs $H_1 : \sigma^2 = \sigma_1^2 > \sigma_0^2$, log-LR:

- A) $\frac{m}{2} \log(\sigma_0^2/\sigma_1^2) + \frac{\sum X_i^2}{2} (1/\sigma_0^2 - 1/\sigma_1^2)$
- B) $\sum X_i^2 (1/\sigma_0^2 - 1/\sigma_1^2)$
- C) $m \log(\sigma_1/\sigma_0)$
- D) $\sum X_i^2/\sigma_0^2$

Answer: A

Trap/Note: $\log \Lambda_m = \sum [\log f(X_i; \sigma_1^2) - \log f(X_i; \sigma_0^2)] = \frac{m}{2} \log(\sigma_0^2/\sigma_1^2) + \frac{\sum X_i^2}{2}(\sigma_0^{-2} - \sigma_1^{-2})$; linear in $\sum X_i^2$

Q354. Decision Theory Shrinkage [HIGH]

Shrinkage estimators are preferred in high dimensions because:

- **A)** They reduce total MSE by trading bias for variance reduction
- **B)** They are always unbiased
- **C)** They attain CRLB
- **D)** They are minimax

Answer: A

Trap/Note: Shrinkage: introduce bias but reduce variance; total MSE decreases; Stein paradox shows this is beneficial for $p \geq 3$; D: James-Stein is minimax but not all shrinkage estimators are

Q355. LRT Poisson Regression [HIGH]

LRT for $H_0: \beta = 0$ in Poisson regression $\log \mu_i = \alpha + \beta x_i$:

- **A)** $-2 \log \Lambda = 2[\ell(\hat{\alpha}, \hat{\beta}) - \ell(\hat{\alpha}_0, 0)] \xrightarrow{d} \chi_1^2$
- **B)** $-2 \log \Lambda \xrightarrow{d} \chi_2^2$
- **C)** $-2 \log \Lambda \xrightarrow{d} F_{1, n-2}$
- **D)** $-2 \log \Lambda \xrightarrow{d} N(0, 1)$

Answer: A

Trap/Note: One restriction ($\beta = 0$): $r = 1$ df; Wilks theorem applies; χ_1^2 ; equivalent to Wald test $(\hat{\beta}/\text{SE})^2$ asymptotically

Q356. SPRT Stopping Time [HIGH]

SPRT stopping time N satisfies:

- **A)** $E_\theta[N] < \infty$ for all θ under regularity conditions
- **B)** $N = n_0$ always (fixed)
- **C)** $N \rightarrow \infty$ as $\alpha, \beta \rightarrow 0$
- **D)** Both A and C

Answer: D

Trap/Note: A: SPRT terminates with probability 1 (Wald); C: as error bounds tighten, more observations needed; both are true

Q357. Decision Theory Empirical Bayes [HIGH]

Empirical Bayes estimates hyperparameters of prior by:

- **A)** Maximizing marginal likelihood $m(x|\lambda) = \int f(x|\theta)\pi(\theta|\lambda)d\theta$
- **B)** Using flat prior
- **C)** Cross-validation

- D) Bootstrap

Answer: A

Trap/Note: Empirical Bayes: estimate λ from data (marginal likelihood or method of moments); then use posterior with estimated prior; James-Stein is an empirical Bayes estimator

Q358. LRT Survival [HIGH]

Log-rank test for comparing two survival curves is:

- A) Asymptotically equivalent to LRT under proportional hazards
- B) Exact LRT for exponential survival
- C) Equivalent to t -test on survival times
- D) Equivalent to Wilcoxon test

Answer: A

Trap/Note: Log-rank test is score test (not LRT) under Cox model; asymptotically equivalent to LRT; exact LRT for exponential gives different statistic; D: Wilcoxon is different weighting

Q359. SPRT Curtailed [HIGH]

Curtailed sampling (stop when outcome is determined) differs from SPRT in that:

- A) Curtailment stops when final decision is certain regardless of remaining observations
- B) Curtailment uses likelihood ratio boundaries
- C) Curtailment is optimal
- D) Curtailment and SPRT are equivalent

Answer: A

Trap/Note: Curtailment: stop when remaining observations cannot change decision; used in clinical trials; SPRT uses probabilistic boundaries; different stopping rules

Q360. Decision Theory Posterior Odds [HIGH]

Posterior odds ratio for H_1 vs H_0 is:

- A) $\frac{P(H_1|X)}{P(H_0|X)} = \frac{P(X|H_1)}{P(X|H_0)} \cdot \frac{P(H_1)}{P(H_0)}$ (Bayes factor $\times$ prior odds)
- B) $P(X|H_1)/P(X|H_0)$ (Bayes factor only)
- C) $P(H_1|X)$ alone
- D) $1 - p$ -value

Answer: A

Trap/Note: Posterior odds = Bayes factor $\times$ prior odds; Bayes factor alone (B) ignores prior; p -value is frequentist (D); A is the complete Bayesian update

Q361. LRT Curved Exponential [HIGH]

For curved exponential family (e.g., $N(\theta, \theta^2)$), Wilks theorem:

- A) Still holds: $-2 \log \Lambda \xrightarrow{d} \chi_r^2$

- B) Fails completely
- C) Holds with modified df
- D) Requires larger sample size

Answer: A

Trap/Note: Wilks theorem holds for curved exponential families under regularity; r = number of restrictions; the curvature affects finite-sample performance but not asymptotic distribution

Q362. SPRT Multihypothesis [HIGH]

For testing among $k > 2$ hypotheses sequentially, SPRT generalizes to:

- A) Sequential probability ratio test with $k(k-1)/2$ pairwise boundaries
- B) Simultaneous SPRT (Armitage)
- C) Repeated significance testing
- D) Group sequential design

Answer: B

Trap/Note: Armitage's simultaneous SPRT: continue until one hypothesis is accepted; boundaries involve all k hypotheses; more complex than pairwise

Q363. Decision Theory Complete Class Normal [HIGH]

For normal location problem with squared error loss, complete class consists of:

- A) All Bayes estimators w.r.t. normal priors and their limits
- B) Only the sample mean
- C) All linear estimators
- D) All unbiased estimators

Answer: A

Trap/Note: For normal location, complete class = Bayes estimators w.r.t. normal priors (conjugate family) and limits; includes James-Stein type estimators; B is too restrictive

Q364. LRT Boundary [HIGH]

When true parameter is on boundary of H_0 (e.g., $H_0 : \theta \geq 0$, true $\theta = 0$), $-2 \log \Lambda \xrightarrow{d}$:

- A) $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ (mixture)
- B) χ_1^2
- C) χ_0^2 (degenerate at 0)
- D) $N(0, 1)$

Answer: A

Trap/Note: Boundary case: $-2 \log \Lambda$ is 0 with prob 1/2 (when $\hat{\theta} < 0$, constrained MLE = 0) and χ_1^2 with prob 1/2; mixture distribution

Q365. SPRT Power One [HIGH]

SPRT has power one property: for any $\theta \neq \theta_0$:

- **A)** $P_\theta(\text{SPRT eventually terminates}) = 1$
- **B)** $P_\theta(\text{reject } H_0) \rightarrow 1 \text{ as } n \rightarrow \infty$
- **C)** Both A and B
- **D)** Neither (SPRT may not terminate)

Answer: C

Trap/Note: SPRT terminates with probability 1 (A); and is consistent (B); both follow from Wald's fundamental identity and the strong law of large numbers applied to the log-likelihood ratio (incomplete in source — trap explanation cuts off here)
